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THE ELIMINATION OF THE PARALLAX IN SATELLITE THEORY

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Abstract. When the perturbation affecting a Keplerian motion is proportional to r^{-n} ($n \geqslant 3$), a canonical transformation of Lie type will convert the system into one in which the perturbation is proportional to r^{-2}. Because it removes parallactic factors, the transformation is called the elimination of the parallax.

In the main problem for the theory of artificial satellites, the elimination of the parallax has been conducted by computer to order 4. The first order in the reduced system may now be integrated in closed form, thereby revealing the fundamental property of the first-order intermediary orbits in line with Newton's Propositio XLIV.

Extension beyond order 1 leads to identify a new class of intermediaries for the main problem in nodal coordinates, namely the radial intermediaries.

The technique of smoothing a perturbation prior to normalizing the perturbed Keplerian system, of which the elimination of the parallax is an instance, is applied to derive the intermediaries in nodal coordinates proposed by Sterne, Garfinkel, Cid–Palacios and Aksnes, and to find the canonical diffeomorphisms which relate them to one another and to the radial intermediaries.

1. Introduction

So much having been published about the main problem in the theory of an artificial satellite, it may be taken as axiomatic that a claim of originality may (and probably will) be met by a statement of the opposite. With all this wealth, empty spaces nevertheless exist where an author can contribute more than an inadvertent duplication or a personal derivation of results already in print. We believe the present communication falls into one of the cracks. The elimination of the parallax is a technique by which a Keplerian system in a central field of force in r^{-n} ($n \geqslant 3$) is converted into a quasi-Keplerian system with a variable angular momentum. To our knowledge, this technique is new to perturbation theories in non-linear and celestial mechanics.

In the theory of artificial satellites, the equations of motion are generally solved analytically in two steps. A canonical transformation eliminates the short period terms depending on the mean anomaly. Then the averaged equations are solved either by numerical integration, as in the semi-analytical theories of Lidov and Cefola (1974) or Wu Lian-Da *et al.* (1978), or by analytical normalization, a second canonical transformation being performed to remove the long-period terms in the argument of perigee and the longitude of the ascending node. Both methods tend generally to underestimate the amount of work required in the course of averaging over the mean anomaly. Yet the practical difficulties met in computing average elements from initial conditions and in evaluating multivariate Fourier series have discouraged engineers from programming analytical solutions for predicting and improving

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positions of satellites in real time. In this context, the elimination of the parallax presents real merits. In a kind of feasibility study to justify a programming effort of long duration, the parallax has been eliminated from the main problem to order 4. Even though they have no realistic use, the results are reported here to let the reader be the judge, at the onset, of our claims to simplicity. Soon after the present manuscript appears in print, there will come a report on what has been accomplished by removing the parallax in a gravity field including the zonal harmonics from J_2 to J_7. It will show how radically the elimination has disintricated the problem.

Undeniably the method imposes the conventional algorithms to start with yet another canonical transformation. However, the procedure amounts to factoring the elimination of the short-period terms into the product of two transformations, each one definitely simpler than the product. The elimination of the parallax removes those short-period variations due to a torsion exercised on the apsidal frame by the perturbations in r^{-n} $(n \geqslant 3)$ and in the argument of latitude. Left in the system are the short-period variations resulting from irregularities in the equation of the centre. The factoring is reminiscent of the parceling of terms made in lunar theory by de Pontécoulant.

Computer programs are in the making to eliminate the parallax in the theory of artificial satellites; they are built by stepwise refinements, each extension closely collating its results with developments obtained by various authors. How else could one guard the computer programs from slipping into error? While this sustained vigilance stimulates to assimilate anterior contributions into the present scheme, it may (and will) trap some readers into commenting that the 'last word has been said long ago in the theory of artificial satellites'. To the contrary, we believe that, stripped of its non-essential complications, the main problem is now amenable to further investigations into its characteristic structures.

The computational complexities induced by the short-period fluctuations have been paid a great deal of attention since they were uncovered by Kozai (1962) while he extended to order 2 the first-order solution developed by Brouwer (1959). One way of mitigating the difficulty consists in breaking the main problem in two parts, a dominant term made of a separable Hamiltonian called the intermediary and a remainder to be regarded as a perturbation affecting the intermediary (Sterne, 1960). In spheroidal coordinates (Vinti, 1959; Kislik, 1961), the intermediaries contain the whole first-order part of the main problem. Apparently this is not the case for the intermediaries in polar coordinates (Sterne, 1957; Garfinkel, 1958, 1959; Cid–Palacios, 1969; Aksnes, 1965, 1966). The deficiency, however, is remedied by reformulating the concept of intermediary. Given a Hamiltonian in the map (q, Q), a *natural* intermediary for H will be defined as a pair (ϕ, J) made of a canonical transformation $\phi : (q, Q) \to (p, P)$ and of a separable Hamiltonian J in the chart (p, P) such that the perturbation $H[q(p, P), Q(p, P)] - J(p, P)$ be of second order. For each intermediary in the sense of Sterne–Garfinkel, there exists an infinitesimal contact transformation which makes it a natural intermediary. This new concept

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is but another instance of the general approach advocated in this paper, whereby normalization of a perturbed Keplerian system is prepared by a canonical transformation designed to achieve specific simplifications. The elimination of the parallax itself may be used as the preparatory transformation leading to yet another family of natural intermediaries in polar coordinates called the *radial* intermediaries.

2. Lie Derivatives in a Keplerian Flow

Consider the vector space $\mathbf{R}^6$ with coordinates (x, y, z, X, Y, Z) and the differential 2-form

$$\omega^2 = \mathrm{d}X \wedge \mathrm{d}x + \mathrm{d}Y \wedge \mathrm{d}y + \mathrm{d}Z \wedge \mathrm{d}z.$$

The phase flow (Arnold, 1978, ch. 8) determined by the Hamiltonian

$$M_0 = \tfrac{1}{2}(X^2 + Y^2 + Z^2) - \mu(x^2 + y^2 + z^2)^{-1/2} \tag{1}$$

on the symplectic manifold $(\mathbf{R}^6, \omega^2)$ is what we call here a *Keplerian flow*. For any function $F(x, y, z, X, Y, Z)$, the Lie derivative of F in the Keplerian flow is the function

$$L_0(F) = (F\,;M_0)$$

defined by the Poisson bracket of F and M_0 in the symplectic manifold $(\mathbf{R}^6, \omega^2)$.

A Whittaker (1904) transformation $(x, y, z, X, Y, Z) \to (r, \theta, v, R, \Theta, N)$ will replace the Cartesian state variables by the polar nodal variables on the symplectic manifold; it will simplify the evaluation of the Lie derivatives $L_0(F)$. Let it be recalled that the rotation of coordinates

$$x = x' \cos v - y' \sin v \cos I,$$

$$y = x' \sin v + y' \cos v \cos I,$$

$$z = y' \sin I$$

extended by the rotation of momenta

$$X = X' \cos v - Y' \sin v \cos I,$$

$$Y = X' \sin v + Y' \cos v \cos I,$$

$$Z = Y' \sin I$$

keeps invariant the differential identity

$$X\,\mathrm{d}x + Y\,\mathrm{d}y + Z\,\mathrm{d}z = X'\,\mathrm{d}x' + Y'\,\mathrm{d}y' + N\,\mathrm{d}v$$

when the angle I is determined by the relations

$$G = x'Y' - y'X' \quad \text{and} \quad \cos I = N/G.$$

Under these conditions, the substitution $(x, y, z, X, Y, Z) \to (x', y', v, X', Y', N)$ defines

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a homogeneous canonical transformation. It is then composed with the canonical extension from Cartesian to polar coordinates $(x', y', X', Y') \to (r, \theta, R, \Theta)$ given by the equations

$$x' = r\cos\theta, \qquad X' = R\cos\theta - \frac{\Theta}{r}\sin\theta,$$

$$y' = r\sin\theta, \qquad Y' = R\sin\theta + \frac{\Theta}{r}\cos\theta.$$

The product transformation is what we call a *Whittaker transformation.*

In the Whittaker chart, the Keplerian Hamiltonian (1) becomes the function

$$M_0 = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r}. \tag{2}$$

The state functions

$$p \equiv p(\Theta) = \Theta^2/\mu, \tag{3}$$

$$C \equiv C(r, \theta, R, \Theta) = \left(\frac{\Theta}{r} - \frac{\Theta}{p}\right)\cos\theta + R\sin\theta, \tag{4_1}$$

$$S \equiv S(r, \theta, R, \Theta) = \left(\frac{\Theta}{r} - \frac{\Theta}{p}\right)\sin\theta - R\cos\theta \tag{4_2}$$

play a prominent role in the course of eliminating the parallax in a perturbed Keplerian flow. The length p is the semi-latus rectum, C the projection of the radial velocity R in the direction of the ascending node and S the projection of R on the perpendicular to the ascending node in the orbital plane. Unusual though they are, the functions C and S came to our attention when we found that they belong to a remarkable set of *ideal* elements for perturbed Keplerian systems (Deprit, 1975). Their partial derivatives

$$\begin{aligned}
&\frac{\partial C}{\partial r} = -\frac{\Theta}{r^2}\cos\theta, && \frac{\partial S}{\partial r} = -\frac{\Theta}{r^2}\sin\theta,\\
&\frac{\partial C}{\partial \theta} = -S, && \frac{\partial S}{\partial \theta} = C,\\
&\frac{\partial C}{\partial R} = \sin\theta, && \frac{\partial S}{\partial R} = -\cos\theta,\\
&\frac{\partial C}{\partial \Theta} = \left(\frac{1}{r} + \frac{1}{p}\right)\cos\theta, && \frac{\partial S}{\partial \Theta} = \left(\frac{1}{r} + \frac{1}{p}\right)\sin\theta,\\
&\frac{\partial C}{\partial \nu} = 0, && \frac{\partial S}{\partial \nu} = 0,\\
&\frac{\partial C}{\partial N} = 0, && \frac{\partial S}{\partial N} = 0
\end{aligned} \tag{5}$$

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with respect to the nodal elements are notably short expressions nowhere singular save at the origin and along linear orbits ($p = 0$).

It has been suggested that dimensionless quantities like

$$C^* \equiv C^*(r, \theta, R, \Theta) = \left(\frac{p}{r} - 1\right)\cos\theta + \frac{Rp}{\Theta}\sin\theta,$$

$$S^* \equiv S^*(r, \theta, R, \Theta) = \left(\frac{p}{r} - 1\right)\sin\theta - \frac{Rp}{\Theta}\cos\theta$$

could be more advantageous than the element C and S which have the dimension of a velocity. But the partial derivatives

$$\frac{\partial C^*}{\partial r} = -\frac{p}{r^2}, \qquad \frac{\partial S^*}{\partial r} = -\frac{p}{r^2},$$

$$\frac{\partial C^*}{\partial \theta} = -S^*, \qquad \frac{\partial S^*}{\partial \theta} = C^*,$$

$$\frac{\partial C^*}{\partial R} = \frac{p}{\Theta}\sin\theta, \qquad \frac{\partial S^*}{\partial R} = -\frac{p}{\Theta}\cos\theta,$$

$$\frac{\partial C^*}{\partial \Theta} = \frac{1}{\Theta}\left[\left(\frac{p}{r} + 1\right)\cos\theta + C^*\right], \qquad \frac{\partial S^*}{\partial \Theta} = \frac{1}{\Theta}\left[\left(\frac{p}{r} + 1\right)\sin\theta + S^*\right],$$

$$\frac{\partial C^*}{\partial v} = 0, \qquad \frac{\partial S^*}{\partial v} = 0,$$

$$\frac{\partial C^*}{\partial N} = 0, \qquad \frac{\partial S^*}{\partial N} = 0,$$

especially those with respect to Θ, are less simple than those for C and S, which is the reason why, in the present context, the state variables C and S have been adopted rather than the more common elements C^* and S^*.

The Lie derivative of the argument of latitude θ in a Keplerian flow results immediately from (2):

$$L_0(\theta) = \frac{\partial M_0}{\partial \Theta} = \frac{\Theta}{r^2}. \tag{6}$$

But the Lie derivatives of C and S in a Keplerian flow are more quickly evaluated in a Delaunay than in a Whittaker chart.

The textbooks in celestial mechanics derive the Delaunay transformation from a complete solution of the Hamilton–Jacobi equation for Keplerian systems. In essence, however, a Delaunay transformation stands on its own as a change of variables in the phase space, irrespective of the dynamical system to which it may be applied. To the same extent, the elimination of the parallax belongs to Geometry

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per se; through the mechanism of Lie series, the construction is applied to whatever Hamiltonian is amenable to it. The mapping

$$\delta : (r, \theta, v, R, \Theta, N) \to (l, g, h, L, G, H)$$

defined implicitly by the equations

$$R = \frac{\partial D}{\partial r}, \qquad \Theta = \frac{\partial D}{\partial \theta}, \qquad N = \frac{\partial D}{\partial v}, \tag{7}$$

$$l = \frac{\partial D}{\partial L}, \qquad g = \frac{\partial D}{\partial G}, \qquad h = \frac{\partial D}{\partial H}$$

which are derived from the function

$$D \equiv D(r, \theta, v, L, G, H) = vH + \theta G + \int_P^r Q^{1/2}\, \mathrm{d}w \tag{8}$$

in which the integrand is the expression

$$Q \equiv Q(w, L, G) = -\frac{\mu^2}{L^2} + 2\frac{\mu}{w} - \frac{G^2}{w^2} \tag{9}$$

and the lower limit $P \equiv P(L, G)$ is a root of the equation

$$Q(P, L, G) = 0 \tag{10}$$

is called a *Delaunay transformation* (Andoyer, 1913) or a *Delaunay map* (Abraham and Marsden, 1978). The name given to this elementary change of state variables should not raise confusion. What recent text-books (see e.g. Brouwer and Clemence, 1961, pp. 544 and 553) call a Delaunay transformation has to do with the removal of a periodic term from a perturbation; its creator dubbed it an *operation* and, since Tisserand (1894), the usage had been to call it a *Delaunay operation* (see e.g. Hill, 1902, p. 187 or Smart, 1953, pp. 185 and 193).

In its developed form, namely

$$R = Q^{1/2}, \qquad \Theta = G, \qquad N = H,$$

$$l = \frac{\mu^2}{L^3} I_1, \qquad g = \theta + G I_2, \qquad h = v,$$

the transformation involves the quadratures

$$I_1 = \int_P^r Q^{-1/2}\, \mathrm{d}w \quad \text{and} \quad I_2 = \int_P^r Q^{-1/2}\, \mathrm{d}\left(\frac{1}{w}\right).$$

Before calculating these integrals, one introduces the quantities $a = a(L)$, $\eta \equiv \eta(L, G)$

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and $e \equiv e(L, G)$ such that

$$L^2 = \mu a, \qquad G = L\eta, \qquad e = (1 - \eta^2)^{1/2}. \tag{11}$$

From these definitions, there results that

$$p = a\eta^2 = a(1 - e^2),$$

$$Q = \frac{\mu}{a}\left[\frac{a(1+e)}{w} - 1\right]\left[1 - \frac{a(1-e)}{w}\right].$$

When $0 \leqslant e < 1$, Equation (10) has two strictly positive roots

$$P = a(1 - e) \quad \text{and} \quad A = a(1 + e).$$

Therefore, to perform the quadrature I_1, one may introduce an eccentric anomaly $E \equiv E(r, L, G)$ such that

$$r = a(1 - e\cos E).$$

On account of the identities

$$A - r = ae(1 + \cos E), \qquad r - P = ae(1 - \cos E),$$

$$Q = \frac{L^2}{r^2}e^2 \sin^2 E, \qquad \mathrm{d}r = ae \sin E \,\mathrm{d}E,$$

the quadrature becomes

$$I_1 = \frac{a^2}{L}(E - e\sin E)$$

and converts Delaunay's equation for l into Kepler's equation

$$l = E - e\sin E.$$

Similarly, to execute the quadrature I_2, one introduces a true anomaly $f \equiv f(r, L, G)$ such that

$$\frac{p}{r} = 1 + e\cos f,$$

then, by reason of the identities

$$\frac{A}{r} - 1 = \frac{e}{1-e}(1 + \cos f), \qquad 1 - \frac{P}{r} = \frac{e}{1+e}(1 - \cos f),$$

$$Q = \frac{G^2}{p^2}e^2 \sin^2 f, \qquad \mathrm{d}\left(\frac{1}{r}\right) = -\frac{e}{p}\sin f \,\mathrm{d}f,$$

the quadrature yields that

$$I_2 = -f/G$$

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and reduces Delaunay's equation for g to the formula

$$\theta = g + f.$$

By definition of the true and eccentric anomalies,

$$r(1 + \cos f) = P(1 + \cos E) \quad \text{and} \quad r(1 - \cos f) = A(1 - \cos E);$$

these formulas lead in turn to the identities

$$r^{1/2} \cos \frac{f}{2} = P^{1/2} \cos \frac{E}{2} \quad \text{and} \quad r^{1/2} \sin \frac{f}{2} = A^{1/2} \sin \frac{E}{2},$$

and finally to the formulas

$$\tan \frac{f}{2} = \left(\frac{1+e}{1-e} \right)^{1/2} \tan \frac{E}{2}, \tag{12}$$

$$r \cos f = a(\cos E - e), \qquad r \sin f = a\eta \sin E.$$

In Delaunay variables, the Keplerian Hamiltonian (2) becomes the function

$$M_0 \equiv M_0(L) = -\frac{1}{2} \frac{\mu^2}{L^2}, \tag{13}$$

while the state variables C and S take the familiar form

$$C \equiv C(g, L, G) = \frac{G}{p} e \cos g, \qquad S \equiv S(g, L, G) = \frac{G}{p} e \sin g. \tag{14}$$

Because neither C nor S depend on the mean anomaly l whereas M_0 depends only on the momentum L, there follows immediately that

$$L_0(C) = 0, \qquad L_0(S) = 0, \quad \text{and} \quad L_0(p) = 0. \tag{15}$$

For any function $F \equiv F(p, C, S, \theta)$, consider the gradient

$$\mathrm{d}F = \partial_1 F \, \mathrm{d}p + \partial_2 F \, \mathrm{d}C + \partial_3 F \, \mathrm{d}S + \partial_4 F \, \mathrm{d}\theta; \tag{16}$$

Suggested by Dieudonné (1960), the notation is meant to stress the major distinction between the coefficient $\partial_4 F$ of $\mathrm{d}\theta$ in the Pfaffian (16) and the partial derivative

$$\frac{\partial F}{\partial \theta} = \partial_4 F + \partial_2 F \frac{\partial C}{\partial \theta} + \partial_3 F \frac{\partial S}{\partial \theta}$$

of $F \equiv F[p(\Theta), C(r, \theta, R, \Theta), S(r, \theta, R, \Theta), \theta]$ in the Whittaker chart. From the chain rule

$$L_0(F) = L_0(p) \, \partial_1 F + L_0(C) \, \partial_2 F + L_0(S) \, \partial_3 F + L_0(\theta) \, \partial_4 F$$

results the fundamental formula

$$L_0(F) = \frac{\Theta}{r^2} \partial_4 F. \tag{17}$$

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In the context of transformations built by Lie series (Deprit, 1969), Equation (17) translates into the following.

PROPOSITION. *For any function $F \equiv F(p, C, S, \theta)$, the quadrature*

$$W \equiv W(p, C, S, \theta) = \int^{\theta} F(p, C, S, w)\, \mathrm{d}w$$

is a formal solution of the partial differential equation

$$L_0(W) = \frac{\Theta}{r^2} F.$$

The elimination of the parallax hinges on this proposition.

3. Elimination of the Parallax

Consider a function M on the symplectic manifold $(\mathbf{R}^6, \omega^2)$. For simplicity, we shall suppose that, in the Whittaker chart, M does not depend on the angle v; later in the paper, the restriction will be relaxed. Assume that M can be expanded as a series of the form

$$M = M_{0,0} + \frac{\Theta^2}{r^2} \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!} M_{n,0} \tag{18}$$

where $M_{0,0}$ is the Keplerian Hamiltonian (2) and, for any $n \geqslant 1$, $M_{n,0}$ denotes a trigonometric polynomial in θ whose coefficients are functions of the state variables p, C, S and N. To avoid unnecessary complexity, we shall suppose that

$$M_{n,0} = \sum_{k \geqslant 0} M_{n,0,k} \cos k\theta. \tag{19}$$

Terms in $\sin k\theta$ will not be considered because they are treated the same way as those in $\cos k\theta$; their presence complicates, but does not in any respect modify, the later construction.

The purpose is to build a canonical transformation $\psi : (r, \theta, v, R, \Theta, N) \to (r', \theta', v', R', \Theta', N')$ to convert (18) into a series

$$M = M'_{0,0} + \frac{\Theta'^2}{r'^2} \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!} M_{0,n} \tag{20}$$

whose coefficients $M_{0,n}$ are functions of the state variables p', C', S' and N'. Throughout this paper, it is agreed that $F'(r', \theta', v', R', \Theta', N')$ designates the function obtained by substituting the primed variables for the original variables in the function $F \equiv F(r, \theta, v, R, \Theta, N)$. Accordingly, $M'_{0,0}$ denotes the Keplerian Hamiltonian (2) in the chart $(r', \theta', v', R', \Theta', N')$; p', C' and S' are derived from (3) and (4) in the same way.

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Although the terms $M'_{n,0}$ are not explicit functions of θ', the transformed series M still depends on θ' implicitly through the state variables C' and S'. The mapping ψ is not intended to remove from M all the terms periodic in θ by taking averages of M over θ.

The dominant term in both series (18) and (20) being the Keplerian Hamiltonian, ψ is chosen to be close to the identity mapping; it will be developed as a formal power series in the small parameter ε. To the first order, ψ is an infinitesimal contact transformation

$$\begin{aligned} r &= r' + \varepsilon \frac{\partial W_1}{\partial R'}, & R &= R' - \varepsilon \frac{\partial W_1}{\partial r'}, \\ \theta &= \theta' + \varepsilon \frac{\partial W_1}{\partial \Theta'}, & \Theta &= \Theta' - \varepsilon \frac{\partial W_1}{\partial \theta'}, \\ v &= v' + \varepsilon \frac{\partial W_1}{\partial N'}, & N &= N' - \varepsilon \frac{\partial W_1}{\partial v'} \end{aligned} \tag{21}$$

derived from a generator

$$W_1 \equiv W_1(r', \theta', v', R', \Theta', N').$$

Subjected to the mapping (21), the first two terms in the series (18) are changed into the expression

$$M'_{0,0} + \varepsilon \left[(M'_{0,0}\,; W_1) + \frac{\Theta'^2}{r'^2} M'_{1,0} \right],$$

the terms of degree higher than 1 being omitted (see e.g. Whittaker, 1904, chap. 11, § 133). Therefore the generator W_1 and the first-order coefficient $M'_{0,1}$ are linked by the partial differential identity

$$L_0(W_1) + \frac{\Theta'^2}{r'^2} M'_{0,1} = \frac{\Theta'^2}{r'^2} M'_{1,0}. \tag{22}$$

By choosing $M'_{0,1} = M'_{1,0,0}$, one reduces (22) to the *Lie equation*

$$L_0(W_1) = \frac{\Theta'^2}{r'^2} \sum_{k \geqslant 1} M'_{1,0,k} \cos k\theta'$$

and, by application of the theorem proved at the end of Section 2, finds that it is satisfied formally by the trigonometric polynomial

$$W_1 = \Theta' \sum_{k \geqslant 1} \frac{1}{k} M'_{1,0,k} \sin k\theta'. \tag{23}$$

Incidentally, with the restriction that v is ignorable in the term $M_{1,0}$, there follows that v' is ignorable in the generator W_1 and, by reason of the transformation equation $N = N'$, that ψ maps each submanifold $N =$ constant into itself.

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Extension of an infinitesimal contact transformation beyond order 1 is a Lie transformation. The mapping is the phase flow

$$\psi^{\varepsilon}:[r'(0), \theta'(0), v'(0), R'(0), \Theta'(0), N'(0)]$$
$$\to [r'(\varepsilon), \theta'(\varepsilon), v'(\varepsilon), R'(\varepsilon), \Theta'(\varepsilon), N'(\varepsilon)]$$

determined by the solution of the Hamiltonian system

$$\begin{aligned} \frac{\mathrm{d}r'}{\mathrm{d}\varepsilon} &= \frac{\partial W}{\partial R'}, & \frac{\mathrm{d}\theta'}{\mathrm{d}\varepsilon} &= \frac{\partial W}{\partial \Theta'}, & \frac{\mathrm{d}v'}{\partial\varepsilon} &= \frac{\partial W}{\partial N'}, \\ \frac{\mathrm{d}R'}{\mathrm{d}\varepsilon} &= -\frac{\partial W}{\partial r'}, & \frac{\mathrm{d}\Theta'}{\mathrm{d}\varepsilon} &= -\frac{\partial W}{\partial \theta'}, & \frac{\mathrm{d}N'}{\mathrm{d}\varepsilon} &= -\frac{\partial W}{\partial v'} \end{aligned} \tag{24}$$

satisfying the initial conditions

$$\begin{aligned} r'(0) &= r, & \theta'(0) &= \theta, & v'(0) &= v, \\ R'(0) &= R, & \Theta'(0) &= \Theta, & N'(0) &= N. \end{aligned} \tag{25}$$

The Hamiltonian W is the generator of the Lie transformation; it is built as a series

$$W = \sum_{n \geqslant 0} \frac{\varepsilon^n}{n!} W_{n+1};$$

The zeroth-order term W_1 is the generator of the infinitesimal contact transformation. The terms $W_n (n \geqslant 2)$ are determined by the condition that they satisfy formally partial differential equations in the form

$$L_0(W_n) + M_{0,n} = \frac{\Theta'^2}{r'^2} \tilde{M}_{0,n}, \tag{26}$$

the functions $\tilde{M}_{0,n}$ in the right-hand member being trigonometric polynomials in θ' of the form (19).

Indeed, assume that the construction has been carried out to order n; the generators $W_k (1 \leqslant k \leqslant n-1)$ have been obtained in a complete and definitive manner as functions of the type

$$W_k \equiv W_k(p', C', S', N', \theta') = \Theta' \sum_{m \geqslant 1} W_{k,m} \sin m\theta'; \tag{27}$$

likewise all elements $M_{i,j} (0 \leqslant i \leqslant n, 0 \leqslant j \leqslant n, 0 \leqslant i+j \leqslant n)$ in the Lie triangle have been obtained as expressions of the form

$$M_{i,j} \equiv M_{i,j}(r', p', C', S', N') = \frac{\Theta'^2}{r'^2} \sum_{k \geqslant 0} M_{i,j,k} \cos k\theta' \tag{28}$$

The induction step at order $n+1$ starts with evaluating the element

$$\tilde{M}_{n,1} = M_{n+1,0} + \sum_{0 \leqslant k \leqslant n} \binom{n}{k} (M_{k,0}; W_{n+1-k})$$

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and continues recursively calculating the provisional elements $(i+j=n+1)$

$$\tilde{M}_{i,j} = \tilde{M}_{i+1,j-1} + \sum_{0 \leqslant k \leqslant i} \binom{i}{k} (M_{k,j-1}\,; W_{i+1-k})$$

along the diagonal of the triangle. It must be verified that the new diagonal is made of terms all in the form (28). But the provisional elements are linear combinations of Poisson brackets of energy-like functions A of the type (28) with action-like functions B of the type (27). Such a Poisson bracket will be evaluated as the expression

$$(A\,;B) = A_1 B_3 - A_3 B_1 + A_2 B_4 - A_4 B_2 \tag{29}$$

with the factors

$$A_1 = r\frac{\partial A}{\partial r}, \qquad A_2 = \frac{\partial A}{\partial \theta}, \qquad A_3 = \frac{\Theta}{p}\frac{\partial A}{\partial R}, \qquad A_4 = \Theta\frac{\partial A}{\partial \Theta}, \tag{30_1}$$

having the dimension of an energy multiplying the dimensionless quantities

$$B_1 = \frac{p}{\Theta}\frac{\partial B}{\partial r}, \qquad B_2 = \frac{1}{\Theta}\frac{\partial B}{\partial \theta}, \qquad B_3 = \frac{1}{r}\frac{\partial B}{\partial R}, \qquad B_4 = \frac{\partial B}{\partial \Theta}. \tag{30_2}$$

The energy-like factors A_1, A_2, A_3, A_4 will immediately come out in the form (23); the dimensionless factors B_1, B_2, B_3, B_4 will be cast in the form of trigonometric polynomials

$$\sum_{k \geqslant 1} B(p', C', S', N') \cos k\theta',$$

possibly by repeatedly multiplying terms in $1/r$ by the 'parallax unity'

$$1 = r * [1/p + (Cp/\Theta)\cos\theta + (Sp/\Theta)\sin\Theta]$$

after the derivations have been carried out. Then it is obvious that the Poisson bracket $(A\,;B)$ will come out in the format (28) of an energy-like term, which proves that the terms in the Lie triangle can be reduced to the standard format.

In the identity (26), one chooses

$$M_{0,n+1} \equiv M_{0,n+1}(p', C', S', N') = M_{0,n+1,0}$$

and, by application of the theorem in Section 2, finds that the Lie equation at order $n+1$

$$L_0(W_{n+1}) = \frac{\Theta'^2}{r'^2} \sum_{k \geqslant 1} M_{0,n+1,k} \cos k\theta'$$

admits the formal solution

$$W_{n+1} = \Theta' \sum_{k \geqslant 1} \frac{1}{k} M_{0,n+1,k} \sin k\theta'.$$

The construction is fit to be programed for an automatic processor of Poisson

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series, especially if the prescriptions given in (29), (30_1) and (30_2) are followed to evaluate Poisson brackets in a Whittaker chart.

The Lie transformation ψ is what we call the *elimination of the parallax*. The terminology will be justified in Section 5.

For the attention of readers who are not familiar with Lie transformations, let it be mentioned that the elimination of the parallax could also be effected using a technique devised by Poincaré (1889, 1891, 1893, 1897) but nowadays named after von Zeipel. To this end, the Delaunay chart is more convenient than the Whittaker chart. To recall, a *Poincaré transformation*

$$\phi : (l, g, h, L, G, H) \to \phi(l', g', h', L', G', H')$$

is defined by the implicit equations

$$L = \frac{\partial S}{\partial l}, \qquad G = \frac{\partial S}{\partial g}, \qquad H = \frac{\partial S}{\partial h},$$

$$l' = \frac{\partial S}{\partial L'}, \qquad g' = \frac{\partial S}{\partial G'}, \qquad h' = \frac{\partial S}{\partial H'}$$

derived from a generator

$$S \equiv S(l, g, h, L', G', H') = L'l + G'g + H'h + \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!} S_n.$$

Following Poincaré, one determines the generator and the new function M^* transformed of M by ϕ from the identity

$$M\left(\frac{\partial S}{\partial l}, \frac{\partial S}{\partial g}, \frac{\partial S}{\partial h}, l, g, h\right) = M^*\left(L', G', H', \frac{\partial S}{\partial L'}, \frac{\partial S}{\partial G'}, \frac{\partial S}{\partial H'}\right).$$

Both members are developed in the powers of the small parameter and the coefficients of like powers of ε are equated. So, at order 0, one obtains that

$$M'_{0,0} = M'_{0,0}(L') = -\frac{\mu^2}{2L'^2}.$$

At the following orders, it must be understood that the polar nodal variables either in the old chart or in the new chart, stand for composite expressions obtained by using the new Delaunay momenta (L', G', H') and the old Delaunay coordinates (l, g, h). With this convention, the Taylor expansion yields that

$$n' \frac{\partial S_1}{\partial l} + \frac{\Theta^2}{r^2} M_{1,0} = \frac{\Theta'^2}{r'^2} M_{0,1}$$

with n standing for the state variable $n \equiv n(L) = \mu^2/L^3$ called the mean motion. The unknown term $M_{0,1}$ being selected as before, therefore the result of S_1 is given by the quadrature

$$S_1 = \frac{\Theta'^2}{n'} \int \mathrm{d}l \sum_{k \geqslant 1} \frac{1}{r^2} M_{1,0,k} \cos k\theta. \tag{31}$$

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It must be recalled that

$$\partial\theta/\partial l = \Theta/nr^2$$

to find that the quadrature in (31) may be performed in closed form, thereby yielding that

$$S_1 = \Theta' \sum_{k \geqslant 1} \frac{1}{k} M_{1,0,k} \sin k\theta.$$

The comments showing that the rules for extending the Lie transformation beyond order 1 are consistent, prove also that the elimination of the parallax can be built in the form of a Poincaré transformation.

The digression is informative in that it shows where the elimination of the parallax differs from the conventional averaging over the mean anomaly; the term separated from the old function M to make the new function M^* may not be an average over the mean anomaly.

4. Torsions for Quasi–Keplerian Systems

The simplest function of type (20) appears to be

$$M = M_{0,0} - \frac{1}{2}\frac{\mu\alpha}{r^2} = \frac{1}{2}\left[R^2 + \frac{\Theta^2}{r^2}\left(1 - \frac{\alpha}{p}\right)\right] - \frac{\mu}{r}; \tag{32}$$

It represents the class of what we call the *quasi-Keplerian* systems. Since the angles v and θ are ignorable, the quasi-Keplerian system has only one degree of freedom, hence it is integrable. Irrespective of whether α is a small parameter or not, the Hamiltonian (32) is separable in the Whittaker chart. A complete solution for the Hamilton–Jacobi equation emanating from (32) is easy to find, but the transformation it generates is time-dependent; rather than composing it with another canonical transformation to eliminate the time, we shall immediately produce a canonical transformation to normalize (32).

The mapping $\delta_1 : (r, \theta, v, R, \Theta, N) \to (l_1, g_1, h_1, L_1, G_1, H_1)$ is defined by the implicit equations derived from the generator

$$D_1 \equiv D_1(r, \theta, v, L_1, G_1, H_1) = vH_1 + \theta(G_1^2 + \mu\alpha)^{1/2} + \int_{P_1}^{r} Q_1^{1/2}\, \mathrm{d}w \tag{33}$$

with the integrand

$$Q_1 \equiv Q_1(w, L_1, G_1) = -\frac{\mu^2}{L_1^2} + 2\frac{\mu}{w} - \frac{G_1^2}{w^2}$$

and the lower limit $P_1 \equiv P_1(L_1, G_1)$ a root of the equation $Q_1(P_1, L_1, G_1) = 0$. Pro-

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ceeding exactly as in the case of the Delaunay transformation, one will introduce the quantities a_1, e_1 and p_1 such that

$$L_1^2 = \mu a_1, \qquad G_1^2 = \mu p_1, \qquad p_1 = a_1(1 - e_1^2),$$

also an eccentric anomaly E_1 and a true anomaly f_1 such that

$$r = a_1(1 - e_1 \cos E_1) \quad \text{and} \quad p_1/r = (1 + e_1 \cos f_1).$$

Eventually one finds that the generalized Delaunay transformation is expressed in closed form by the implicit equations

$$R = Q_1^{1/2} = \frac{G_1}{p_1} e_1 \sin f_1 = \frac{L_1}{r} e_1 \sin E_1, \tag{34_1}$$

$$\Theta = (G_1^2 + \mu\alpha)^{1/2}, \tag{34_2}$$

$$N = H_1, \tag{34_3}$$

$$l_1 = E_1 - e_1 \sin E_1, \tag{34_4}$$

$$\theta = \left(1 + \frac{\mu\alpha}{G_1^2}\right)^{1/2} (g_1 + f_1), \tag{34_5}$$

$$h_1 = v. \tag{34_6}$$

In view of Equations (34_1) and (34_2), the quasi-Keplerian Hamiltonian (32) becomes the Keplerian system $M = -\frac{1}{2}(\mu^2/L_1^2)$.

The quasi-Delaunay transformation is in essence the geometric construction outlined by Newton in *Propositio XLIV, Sectio IX, Liber I* of the *Philosophiae naturalis Principia mathematica* (as quoted from Koyré and Cohen, 1972). Imagine in the chart $(l_1, g_1, h_1, L_1, G_1, H_1)$ a phantom satellite S_1 in Keplerian revolution with period T_1 on a fixed ellipse (semi-major axis a_1, eccentricity e_1, argument of perigee g_1). Let t_0 be the time at which the phantom makes a passage at perigee; assume that the (real) satellite S occupies the same position at t_0. As S_1 pursues its course on the ellipse, S enters a rosette-like orbit, either surging ($\alpha > 0$) ahead of the phantom or trailing it ($-p_1 < \alpha < 0$). At any given time, S and S_1 are at the same distance from the centre of the Earth and they have the same radial velocity; but their relative positions and angular velocities are continuously increased or decreased in the constant ratio

$$\frac{\Theta}{G_1} = \frac{\theta}{g_1 + f_1} = \left(1 + \frac{\mu\alpha}{G_1^2}\right)^{1/2} = \left(1 - \frac{\mu\alpha}{\Theta^2}\right)^{-1/2}. \tag{35}$$

The successive apogees of S form a regular polygon inscribed in the circle of radius $a_1(1 + e_1)$ at the centre of the Earth; likewise, its perigees make a regular polygon in a concentric circle of radius $a_1(1 - e_1)$. Every time the phantom S_1 completes a revolution, the perigee and the apogee of the real satellite jump forward or backward

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by an angle $\Delta = 2\pi[(1 + \mu\alpha/G_1^2)^{1/2} - 1]$. It is incorrect to read the formula as meaning that the perigee of S rotates continuously at the constant rate

$$\frac{\Delta}{T_1} = n_1\left[\left(1 + \frac{\mu\alpha}{G_1^2}\right)^{1/2} - 1\right] \tag{36}$$

per anomalistic period T_1, for the rosette-like orbit of S is not generated by an osculating ellipse in continuous deformation.

The generalized Delaunay transformation determined by the function

$$D_2 \equiv D_2(r, \theta, v, L_2, G_2, H_2) = vH_2 + \theta G_2 + \int_{P_2}^{r} Q_2^{1/2}\, \mathrm{d}w \tag{37}$$

with the integrand

$$Q_2 \equiv Q(w, L_2, G_2) = -\frac{\mu^2}{L_2^2} + 2\frac{\mu}{w} - \frac{G_2^2 - \mu\alpha}{w^2}$$

will also convert the quasi-Keplerian system (32) into a Keplerian system in the chart $(l_2, g_2, h_2, L_2, G_2, H_2)$. On the one hand, the elements l_2, h_2, L_2, H_2 are identical to the elements l_1, h_1, L_1, H_1 respectively; on the other, $G_2 = \Theta = (G_1^2 + \mu\alpha)^{1/2}$ and $\theta = g_2 + (1 + \mu\alpha/G_1^2) f_1$; whereas g_1 is the argument of *the* perigee for the phantom, g_2 is the argument of *one* of the perigees for the satellite S.

This form of Newton's geometric construction makes up the basic step in the programs VARPAR and NODE designed (Claus and Lubowe, 1963; Lubowe, 1964 and 1966) for communication satellites in the class TELSTAR. From a set of initial values in the Whittaker chart, one calculates the elements of the phantom's elliptic orbit and the argument of the closest past perigee of the satellite. For a time span equal to the phantom's anomalistic period T_1, one computes positions and velocities of the phantom S_1 and generates from them the positions and velocities of the satellite by applying Newton's construction. At the end of the period, there occurs a sort of *rectification*: from the state reached by the satellite S, the second-order perturbations averaged over T_1 being now taken into account, new elliptic elements are calculated for a phantom, and the procedure is restarted.

Rather than resorting to Delaunay-like variables for reducing the quasi-Keplerian system (32) to a Keplerian system, one can proceed directly in terms of Whittaker elements. To this end is introduced a new class of canonical transformations. A mapping $\tau : (r, \theta, v, R, \Theta, N) \to (r_1, \theta_1, v_1, R_1, \Theta_1, N_1)$ is called a *torsion* if it is defined by the implicit equations derived from a generator of the type $T \equiv T(r_1, \theta_1, v_1, R, \Theta, N) = r_1 R + \theta_1 \Theta\Phi(\Theta, N) + v_1 N\Psi(\Theta, N)$. A torsion owes its name to the fact that it does not modify the radial variables (distance r and radial velocity R).

The torsion defined by

$$T = r_1 R + v_1 N + \theta_1 \Theta\left(1 - \frac{\mu\alpha}{\Theta^2}\right)^{1/2}. \tag{38}$$

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reduces the quasi-Keplerian system (32) to a Keplerian problem in the chart $(r_1, \theta_1, v_1, R_1, \Theta_1, N_1)$. The transformation equations

$$\theta = \frac{\partial T}{\partial \Theta} = \theta_1 \left(1 - \frac{\mu}{p}\right)^{1/2}, \qquad \Theta_1 = \frac{\partial T}{\partial \theta_1} = \Theta \left(1 - \frac{\mu}{p}\right)^{1/2}$$

are equivalent to formula (35). On the one hand, the generalized Delaunay transformation δ_1 is the composition of the torsion generated by (38) and of a Delaunay transformation $(r_1, \theta_1, v_1, R_1, \Theta_1, N_1) \to (l_1, g_1, h_1, L_1, G_1, H_1)$. On the other hand, the torsion (38) is more general than δ_1 in that it does not restrict the phantom to elliptic orbits – it excludes only linear orbits.

In the cases where α is a small parameter, conventional perturbation algorithms may be applied to the Hamiltonian (32) taken as a power series in α truncated below order 1 (see e.g. Chazy, 1953, p. 228). Averaging over the mean anomaly amounts to extending an infinitesimal contact transformation

$$r = r' + \frac{\partial W}{\partial R'}, \qquad \theta = \theta' + \frac{\partial W}{\partial \Theta'}, \qquad v = v',$$

$$R = R' - \frac{\partial W}{\partial r'}, \qquad \Theta = \Theta', \qquad N = N'$$

whose generator is

$$W = -\tfrac{1}{2} G' \frac{\alpha}{p'} (f' - l). \tag{39}$$

The averaged Hamiltonian then becomes

$$M' = -\tfrac{1}{2} \frac{\mu}{a'} \left(1 + \frac{\alpha}{p'} \eta'\right). \tag{40}$$

The geometric model behind the analytical approximation is a familiar one. In the Delaunay chart (l', g', h', L', G', H'), the actions L', G' and H' are integrals while the angles l', g' and h' are linear functions of the time with the mean motions

$$\dot{l}' = \frac{\partial M'}{\partial L'} = n' \left(1 + \frac{3}{2} \frac{\alpha}{p'} \eta'\right), \qquad \dot{g}' = \frac{\partial M'}{\partial G'} = \tfrac{1}{2} n' \frac{\alpha}{p'}, \qquad \dot{h}' = \frac{\partial M'}{\partial H'} = 0. \tag{41}$$

The reference orbit in the Whittaker chart $(r', \theta', v', R', \Theta', N')$ is regarded as an ellipse of constant semi-major axis a' and eccentricity e' with a perigee rotating at a slow uniform rate $\dot{g}'$; the secular variation in l' induces a tangential drift from a proper Keplerian revolution. At the top level in the Whittaker chart $(r, \theta, v, R, \Theta, N)$, there appear short-period corrections affecting not only the position angle θ but also the radial elements r and R:

$$r = r' \left\{ 1 - \frac{\alpha}{p'} \left[\eta' + \frac{1}{2} \frac{1}{1 + \eta'} \frac{p'}{r'} \left(\frac{p'}{r'} - 1 \right) \right] \right\},$$

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$$R = R'\left[1 + \frac{1}{2}\frac{\alpha}{p'}\left(\eta' + \frac{1}{1+\eta'}\frac{p'^2}{r'^2}\right)\right],$$

$$\theta = \theta' + \frac{1}{2}\frac{\alpha}{p'}\left[f' - l' + \frac{1}{1+\eta'}\left(\frac{p'}{r'} + 1\right)e' \sin f'\right].$$

The model is one often met in the world of artifacts produced by the methods of averaging over the mean anomaly; extension to higher orders comes easily since it can be proved that every step of the induction will yield the generator as a purely periodic function of short period expressed in closed form with no development in the powers of the eccentricity e'. Yet, familiar as it may be, the model is definitely foreign to the exact solution which, to recall, is a rosette developing from a torsion applied to a proper Keplerian orbit. At first sight, the only contact between the artifact and the exact model is formula (41) for the secular variation of the average perigee which is the development of the leap rate (36) as a Taylor series in the powers of α truncated below order 1.

For an approximation scheme that adheres closely to the structures of the exact solution, we propose the infinitesimal contact transformation generated by

$$W = -\tfrac{1}{2}\,\Theta'\frac{\alpha}{p'}\theta'. \tag{42}$$

It results in the transformation equations

$$r = r' + \frac{\partial W}{\partial R'} = r', \qquad \theta = \theta' + \frac{\partial W}{\partial \Theta'} = \theta'\left(1 + \frac{1}{2}\frac{\alpha}{p'}\right),$$

$$R = R' - \frac{\partial W}{\partial r'} = R', \qquad \Theta = \Theta' - \frac{\partial W}{\partial \theta'} = \Theta'\left(1 + \frac{1}{2}\frac{\alpha}{p'}\right).$$

They show that the infinitesimal contact transformation does in the small about $\alpha = 0$ what the finite torsion (35) does in the exact solution: (i) it does not affect the radial elements ; (ii) it induces an infinitesimal torsion

$$\frac{\theta}{\theta'} = \frac{\Theta}{\Theta'} = 1 + \frac{1}{2}\frac{\alpha}{p'}$$

tangent to the finite torsion ; (iii) it converts the quasi-Keplerian Hamiltonian (32) into precisely a Keplerian Hamiltonian so that the reference orbit remains a constant ellipse with *fixed* perigee. As a matter of fact, it can be extended beyond order 1 : if it is developed as a Lie transformation in the powers of α, then the generator is obtained by applying a procedure due to Shniad (1970) to the torsion generator (38). That the generator (42) contains θ as a factor is not objectionable ; on the contrary it indicates that the transformation does not proceed by taking averages over the mean anomaly or even the argument of the latitude θ, but that it applies a torsion to a proper Keplerian motion.

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5. The Main Problem in the Theory of Artificial Satellites

The tools prepared in Sections 3 and 4 will now be applied to the theory of artificial satellites.

Let $(\mathbf{I}, \mathbf{J}, \mathbf{K})$ be the orthonormal base for a Cartesian frame (U) fixed in space at the centre of the Earth; also let $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ be the orthonormal base for a Cartesian frame (B) fixed in the Earth at its centre. Let $\mathbf{x}$ be the position of the satellite, taken as a massless point, with respect to the centre of the Earth; set

$$\mathbf{x} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \qquad x = r\cos\beta\cos\lambda,$$
$$y = r\cos\beta\sin\lambda, \qquad z = r\sin\beta.$$

The velocity of the satellite relative to the fixed frame (U) is the vector

$$\mathbf{X} = \dot{\mathbf{x}} + \boldsymbol{\omega} \times \mathbf{x}, \tag{43}$$

with $\boldsymbol{\omega}$ denoting the angular velocity of rotation of the Earth frame (B) relative to the space frame (U) and $\dot{\mathbf{x}}$ standing for the velocity of the satellite in the moving frame (B), that is to say for the vector $\dot{\mathbf{x}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k}$. The potential of the gravity field of the Earth is usually expressed in terms of the spherical coordinates (r, λ, β) by a series of spherical harmonics

$$V = -\frac{\mu}{r}\sum_{n\geqslant 0}\left(\frac{\alpha}{r}\right)^n \sum_{0\leqslant m\leqslant n}(C_n^m\cos m\lambda + S_n^m\sin m\lambda)P_n^m(\sin\beta). \tag{44}$$

where the functions p_n^m are the associated Legendre polynomials of degree n and order m. The physical parameters μ and α are respectively the Earth's Keplerian constant, i.e. the product of the mass of the Earth by the constant of gravitation, and the 'equatorial' radius of the Earth. The triangular matrices C_n^m $(n \geqslant 0, 0 \leqslant m \leqslant n)$ and S_n^m $(n \geqslant 1, 0 \leqslant m \leqslant n)$ are made of dimensionless elements. Evidently $C_0^0 = 1$; also $C_1^0 = C_1^1 = S_1^1 = 0$ since the Earth's barycentre is the origin of coordinates.

The theory of artificial satellites is the analysis of the dynamical system represented by the Lagrangian

$$K = \tfrac{1}{2}(\mathbf{X}\cdot\mathbf{X}) + V. \tag{45}$$

The Legendre transformation $\dot{\mathbf{x}} \to \mathbf{X}$ defined by the relations

$$X = (\partial K/\partial\dot{x}) = \dot{x} + \boldsymbol{\omega} \times \mathbf{x}\cdot\mathbf{i},$$
$$Y = (\partial K/\partial\dot{y}) = \dot{y} + \boldsymbol{\omega} \times \mathbf{x}\cdot\mathbf{j},$$
$$Z = (\partial K/\partial\dot{z}) = \dot{z} + \boldsymbol{\omega} \times \mathbf{x}\cdot\mathbf{k},$$

associates with (45) the Hamiltonian

$$M = \mathbf{X}\cdot\dot{\mathbf{x}} - K = \tfrac{1}{2}(\mathbf{X}\cdot\mathbf{X}) - \boldsymbol{\omega}\cdot(\mathbf{X}\times\mathbf{x}) + V. \tag{46}$$

The reader should be aware of a disparity in meaning between the coordinates

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(x, y, z) and their conjugate momenta (X, Y, Z). Whereas (x, y, z) are the components of the position $\mathbf{x}$ in the frame (B), (X, Y, Z) are the components in the same frame of the velocity $\mathbf{X}$ relative to the fixed frame (U).

The Hamiltonian (46) depends on the time only through the angular velocity $\boldsymbol{\omega}$. From the onset the theory will be simplified by supposing that $\boldsymbol{\omega} = n_e\mathbf{k}$ and $n_e =$ constant; under these assumptions (46) is a conservative Hamiltonian, and the dynamical system admits the energy integral $M =$ constant.

A Whittaker transformation $(x, y, z, X, Y, Z) \to (r, \theta, v, R, \Theta, N)$ is performed to convert (46) into a function of the form

$$M = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - n_e N + V(r, \theta, v, \Theta, N).$$

The auxiliary angle I measures the inclination of the satellite's orbital plane above the Earth's equatorial plane $(\mathbf{i}, \mathbf{j})$ while v is the longitude of the ascending node reckoned from the rotating axis of inertia $\mathbf{i}$ in the direct sense from East to West (not the right ascension of the node counted from $\mathbf{I}$).

The practical problem of carrying out the substitutions

$$\begin{aligned} \cos(\lambda - v)\cos\beta &= \cos\theta, \\ \sin(\lambda - v)\cos\beta &= \sin\theta\cos I, \\ \sin\beta &= \sin\theta\sin I \end{aligned}$$

in the potential (44) will not be tackled here. For we are planning an approach by stepwise refinements in which the programming leapfrogs from a simplified version of the problem to the next phase in complexity. To begin with, the system is reduced to one with only two degrees of freedom by omitting the tesseral harmonics C_n^m and S_n^m $(n \geqslant 2, 1 \leqslant m \leqslant n)$ in (44). In the resulting Hamiltonian

$$M = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - n_e N + V(r, \theta, v, \Theta, N),$$

the longitude v is ignorable, hence the polar component N of the angular momentum is an integral.

For conciseness in notations, set $\varepsilon = C_2^0$, $c = \cos I$, $s = \sin I$.

From the programming standpoint, there arise now two problems: (i) the expansion of V in Fourier series of θ, and (ii) the elimination of the parallax from the result. A forthcoming communication will explain how the expansion has been dealt with; at present we content ourselves with solving the second problem. In the case of the Earth, the second zonal harmonic coefficient ε is about 100 times bigger in absolute value than any other zonal harmonic. For the time being, the Hamiltonian may thus be reduced even further to the function

$$M = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - n_e N - \frac{\mu}{r}\left[1 + \varepsilon\left(\frac{\alpha}{r}\right)^2 P_2(s\sin\theta)\right]; \tag{47}$$

it defines what we call the *main problem* in the theory of artificial satellites.

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The development of the Legendre polynomial P_2 is a trivial operation :

$$P_2(s \sin \theta) = -(\tfrac{1}{2} - \tfrac{3}{4}s^2) - \tfrac{3}{4}s^2 \cos 2\theta. \tag{48}$$

With the notations introduced in Section 2,

$$\frac{\mu}{r}\left(\frac{\alpha}{r}\right)^2 = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2\left(\frac{p}{r}\right) = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2\left(1 + \frac{Cp}{\Theta}\cos\theta + \frac{Sp}{\Theta}\sin\theta\right). \tag{49}$$

Here is the place where the parallax α/r is formally converted into the ratio α/p. The algorithm outlined in Section 3 purports to remove the sum of periodic terms by which the parallax has been replaced, hence the name given to the transformation.

Substituting the expressions (48) and (49) in the Hamiltonian (47) yields

$$\begin{aligned} M &= M_{0,0} + \varepsilon \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 M_{1,0}, \\ M_{0,0} &= \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r} - n_e N, \\ M_{1,0} &= (\tfrac{1}{2} - \tfrac{3}{4}s^2) + \tfrac{3}{4}s^2 \cos 2\theta + \frac{Cp}{\Theta}\left[(\tfrac{1}{2} - \tfrac{3}{8}s^2)\cos\theta + \tfrac{3}{8}s^2 \cos 3\theta\right] \\ &\quad + \frac{Sp}{\Theta}[(\tfrac{1}{2} - \tfrac{9}{8}s^2)\sin\theta + \tfrac{3}{8}s^2 \sin 3\theta]. \end{aligned} \tag{50}$$

The expression shows that the main problem (47) belongs to the class of Hamiltonians defined in Equation (18). The elimination of the parallax will convert it into a series

$$M = M_{0,0} + \frac{\Theta^2}{r^2}\sum_{n \geqslant 1}\frac{\varepsilon^n}{n!}\left(\frac{\alpha}{p}\right)^{2n}\left[\sum_{0 \leqslant j \leqslant n/2}\sum_{0 \leqslant k \leqslant j}\left(\frac{Sp}{\Theta}\right)^{2k}\left(\frac{Cp}{\Theta}\right)^{2(j-k)} M_{n,j,k}\right]; \tag{51}$$

whose coefficients $M_{n,j,k}$ are the polynomials in s^2 listed in Table I. The context makes it clear that the Whittaker variables used there are the new nodal polar variables determined by the transformation. It may be in order to emphasize that the developments involve no expansion in the powers of either e or s : the mapping is formally consistent over the ranges $0 \leqslant e < \infty$ and $0 \leqslant I \leqslant \pi$. Note also that the transformation applies everywhere in the phase space save on the submanifold $\Theta = 0$.

Indications about the complexity of the computer program have been given elsewhere (Deprit, 1978). The calculations have been executed in three independent ways. By means of his own drastic revision of MAO (Rom, 1969) now written in PL/I for the IBM Optimizing Compiler (Version 1, release 3.0), the author completed the calculations to order 4 on the Amdahl 470–6 at the University of Cincinnati by Christmas 1978. Then, in January 1979, with an updated version of a processor of Poisson series composed in Fortran by Dasenbrock (1973), Dr K. T. Alfriend wrote a "quick dirty" program for the array-processor TI Advanced Scientific Computer

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TABLE I

Coefficients in the main problem for the theory of artificial satellites after elimination of the parallax

$M_{1,0,0} =$	$\frac{1}{2}$	$-\frac{3}{4}s^2$			
$M_{2,0,0} =$	$-\frac{5}{4}$	$+\frac{21}{8}s^2$	$-\frac{21}{16}s^4$		
$M_{2,1,0} =$	$-\frac{3}{8}$	$+\frac{27}{16}s^2$	$-\frac{75}{64}s^4$		
$M_{2,0,1} =$	$-\frac{3}{8}$	$-\frac{15}{16}s^2$	$+\frac{105}{64}s^4$		
$M_{3,0,0} =$	$\frac{39}{4}$	$-\frac{567}{16}s^2$	$+\frac{2961}{64}s^4$	$-\frac{315}{16}s^6$	
$M_{3,1,0} =$	$\frac{87}{16}$	$-\frac{855}{32}s^2$	$+\frac{6345}{128}s^4$	$-\frac{13\,725}{512}s^6$	
$M_{3,0,1} =$	$\frac{87}{16}$	$-\frac{819}{32}s^2$	$+\frac{7281}{128}s^4$	$-\frac{18\,855}{512}s^6$	
$M_{4,0,0} =$	$-\frac{501}{4}$	$+\frac{18\,909}{32}s^2$	$-\frac{131\,157}{128}s^4$	$+\frac{50\,049}{64}s^6$	$-\frac{13\,815}{64}s^8$
$M_{4,1,0} =$	$-\frac{3633}{32}$	$+\frac{64\,467}{64}s^2$	$-\frac{578\,541}{256}s^4$	$+\frac{1\,969\,719}{1024}s^6$	$-\frac{2\,123\,955}{4096}s^8$
$M_{4,0,1} =$	$-\frac{3633}{32}$	$-\frac{16\,623}{64}s^2$	$+\frac{623\,559}{256}s^4$	$-\frac{3942\,663}{1024}s^6$	$+\frac{7\,316\,505}{4096}s^8$
$M_{4,2,0} =$	$-\frac{783}{128}$	$+\frac{9369}{256}s^2$	$-\frac{144\,585}{1024}s^4$	$+\frac{434\,619}{2048}s^6$	$-\frac{1\,557\,765}{16384}s^8$
$M_{4,1,1} =$	$-\frac{783}{64}$	$+\frac{13\,905}{128}s^2$	$-\frac{325\,161}{512}s^4$	$+\frac{533\,385}{512}s^6$	$-\frac{4\,087\,665}{8192}s^8$
$M_{4,0,2} =$	$-\frac{783}{128}$	$+\frac{18\,441}{256}s^2$	$-\frac{204\,849}{1024}s^4$	$+\frac{333\,801}{2048}s^6$	$-\frac{468\,045}{16\,384}s^8$

at the Naval Research Laboratory to satisfy himself that the author's results to order 3 were indeed correct. Finally, using the same software package and the same hardware, Dr S. L. Coffey wrote a fully structured program in *Ratfor* (Kernighan and Plauger, 1976) wherewith, by June 1979, he reproduced the author's results term for term to order 4. (N.B. All three programs proceeding in rational arithmetic, the numerical coefficients in the series are independent of the hardware characteristics and of the conversion subroutines in the compilers.)

The generator of the transformation is a series of the form

$$W = \Theta \sum_{n \geqslant 0} \frac{\varepsilon^n}{n!} \left(\frac{\alpha}{p}\right)^{2(n+1)} W_{n+1}(p, C, S, s^2, \theta),$$

its coefficients being trigonometric polynomials in θ. For the first order,

$$W_1 = \tfrac{3}{8}s^2 \sin 2\theta + \frac{Cp}{\Theta}[(\tfrac{1}{2} - \tfrac{3}{8}s^2)\sin\theta + \tfrac{1}{8}s^2 \sin 3\theta]$$
$$- \frac{Sp}{\Theta}[(\tfrac{1}{2} - \tfrac{9}{8}s^2)\cos\theta + \tfrac{1}{8}s^2 \cos 3\theta]; \tag{52}$$

needless to say, the result could be read immediately from (50). For the second order,

$$W_2 = (\tfrac{5}{4} - \tfrac{21}{16}s^2)s^2 \sin 2\theta + \tfrac{3}{64}s^4 \sin 4\theta + \frac{Cp}{\Theta}[-(2 - \tfrac{131}{16}s^2 + \tfrac{195}{32}s^4)\sin\theta$$
$$- (\tfrac{5}{16} - \tfrac{29}{64}s^2)s^2 \sin 3\theta + \tfrac{3}{64}s^4 \sin 5\theta] + \frac{Sp}{\Theta}[(2 + \tfrac{23}{16}s^2 - \tfrac{141}{32}s^4)\cos\theta$$

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$$+(\tfrac{5}{16}-\tfrac{35}{64}s^2)s^2\cos 3\theta-\tfrac{3}{64}s^4\cos 5\theta]+\frac{C^2p^2}{\Theta^2}[-(\tfrac{3}{16}-\tfrac{3}{8}s^2+\tfrac{9}{256}s^4)\sin 2\theta$$

$$-(\tfrac{15}{64}-\tfrac{75}{256}s^2)s^2\sin 4\theta+\tfrac{3}{256}s^4\sin 6\theta]+\frac{CSp^2}{\Theta^2}[(\tfrac{3}{8}-\tfrac{3}{8}s^2-\tfrac{45}{128}s^4)\cos 2\theta$$

$$+(\tfrac{15}{32}-\tfrac{39}{64}s^2)s^2\cos 4\theta-\tfrac{3}{128}s^4\cos 6\theta]+\frac{S^2p^2}{\Theta^2}[(\tfrac{3}{16}-\tfrac{39}{256}s^4)\sin 2\theta$$

$$+(\tfrac{15}{64}-\tfrac{81}{256}s^2)s^2\sin 4\theta-\tfrac{3}{256}s^4\sin 6\theta].$$

The generators W_3 and W_4 are too long to be printed here. With the first two terms of W, the reader already has enough information against which to check a program of his own to conduct an elimination of the parallax for the theory of artificial satellites.

The conversion equations at the first order are easily derived by hand :

$$\frac{r-r'}{p'}=\varepsilon\left(\frac{\alpha}{p'}\right)^2(\tfrac{1}{2}-\tfrac{3}{4}s'^2-\tfrac{1}{4}s'^2\cos 2\theta'),$$

$$\theta-\theta'=\varepsilon\left(\frac{\alpha}{p'}\right)^2\{\tfrac{3}{4}-\tfrac{7}{8}s'^2-\frac{C'p'}{\Theta'}[(\tfrac{7}{4}-\tfrac{7}{4}s'^2)\sin\theta'-c'^2\sin 3\theta']$$

$$+\frac{S'p'}{\Theta'}[(\tfrac{13}{4}-\tfrac{17}{4}s'^2)\cos\theta'-c'^2\cos 3\theta']\},$$

$$v-v'=\varepsilon\left(\frac{\alpha}{p'}\right)^2c'\left[-\tfrac{3}{4}\sin 2\theta'+\frac{C'p'}{\Theta'}(\tfrac{3}{4}\sin\theta'-\tfrac{1}{4}\sin 3\theta')\right.$$

$$\left.-\frac{S'p'}{\Theta'}(\tfrac{9}{4}\cos\theta'-\tfrac{1}{4}\cos 3\theta')\right],$$

$$\frac{R-R'}{\Theta'/p'}=\tfrac{1}{2}\varepsilon\left(\frac{\alpha}{p'}\right)^2\left(\frac{p'}{r'}\right)^2s'^2\sin 2\theta',$$

$$\frac{\Theta-\Theta'}{\Theta'}=\varepsilon\left(\frac{\alpha}{p'}\right)^2\left\{\tfrac{1}{2}-\tfrac{3}{4}s'^2-\frac{p'}{r'}[(\tfrac{1}{2}-\tfrac{3}{4}s'^2)+\tfrac{3}{4}s'^2\cos 2\theta']\right\},$$

$$N=N'.$$

The explicit transformations from old to new variables (to set the initial conditions) and, conversely, from the new variables to the old ones (to calculate positions and velocities at any given time) have been derived by computer to order 3 by Dr Coffey (1980).

6. Brouwer's Theory After the Parallax has been Eliminated

Elimination of the parallax α/r simplifies but does not solve the main problem in the theory of artificial satellites. Past that preparatory step, several approaches toward a solution are open.

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It is certain that the method advocated by Brouwer is applicable: an averaging over the mean anomaly to remove the short-period terms, followed by an averaging over the argument of the perigee to complete the normalization. Eliminating the short-period terms is strikingly shorter for the new Hamiltonian (51) than for the original one (47). That Brouwer's averaged Hamiltonian is recovered term for term proves that the intrinsic characteristics of the phase space for the main problem have not been altered by removing the parallax.

The task at hand is to build a Lie transformation $(l, g, h, L, G, H) \to (l', g', h', L', G', H')$ for converting the Hamiltonian

$$M = M_{0,0} + \varepsilon M_{1,0} + \frac{\varepsilon^2}{2!} M_{2,0} + O(\varepsilon^3),$$

$$M_{0,0} = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r} - n_e N,$$

$$M_{1,0} = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s^2),$$

$$M_{2,0} = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^4 [-\tfrac{13}{8} + 3s^2 - \tfrac{69}{64}s^4 + (\tfrac{3}{8} - \tfrac{3}{8}s^2 + \tfrac{15}{64}s^4)\eta^2 + (\tfrac{21}{16} - \tfrac{45}{32}s^2)s^2e^2 \cos 2g]$$

into a series

$$M' \equiv M'(g', h', L', G', H') = M'_{0,0} + \varepsilon M_{0,1} + \frac{\varepsilon^2}{2!} M_{0,2} + O(\varepsilon^3)$$

in which l' is ignorable. Note that the term $M_{2,0}$ is a re-arrangement of the second-order term in (51) to revert to variables more familiar than the elements C and S defined in Section 2.

The averaging is built in the manner of a Lie rather than a Poincaré transformation. The reason is not that the generator of the Poincaré transformation is longer or more intricate than that of a Lie transformation; the fact is they are comparable in size and composition. But, as was observed by Hori (1966), the invariance of Poisson brackets with respect to the group of canonical transformations offers ways of simplifying the operations in building a Lie transformation.

Since the longitude of the node h is ignorable in M, Brouwer requires that the corresponding variable h' be ignorable in the generator of the Lie transformation.

At order 1, the Lie equation is

$$n' \frac{\partial W_1}{\partial l'} - n_e \frac{\partial W_1}{\partial h'} + M_{0,1} = M'_{1,0}.$$

For the new term $M_{0,1}$, one selects the average of $M'_{1,0}$ over the mean anomaly;

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then the generator is given by a quadrature in l'. The calculations are straightforward; they hinge on the relation

$$\int \frac{\mathrm{d}l}{r^2} = \frac{n}{G} f.$$

The results are:

$$M_{0,1} = n'^2 a'^2 \left(\frac{\alpha}{p'}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s'^2)\eta' = \eta' G' \left(\frac{\alpha}{p'}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s'^2),$$

$$W_1 = G' \left(\frac{\alpha}{p'}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s'^2)\phi',$$

where $\phi' = f' - l'$ is the equation of the centre.

The second order begins with evaluating the provisional element $\tilde{M}_{1,1} = M'_{2,0} + (M'_{1,0}; W_1)$ in the Lie triangle. The Poisson bracket is more easily computed in the Whittaker than in the Delaunay chart. From the partial derivatives

$$r' \frac{\partial M'_{1,0}}{\partial r'} = -2 \frac{G'^2}{r'^2} \left(\frac{\alpha}{p'}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s'^2),$$

$$\frac{1}{r'} \frac{\partial W_1}{\partial R'} = \left(\frac{\alpha}{p'}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s'^2)(\tfrac{1}{2} + \tfrac{3}{2}\eta' + \beta' e' \cos f' + \tfrac{1}{2}\beta' e'^2 \cos 2f')$$

with $\beta' = (1 + \eta')^{-1}$, it is readily found that

$$\begin{aligned}
\tilde{M}_{1,1} &= M'_{2,0} + \left(r' \frac{\partial M'_{1,0}}{\partial r'}\right)\left(\frac{1}{r'} \frac{\partial W_1}{\partial R'}\right) \\
&= \frac{G'^2}{r'^2} \left(\frac{\alpha}{p'}\right)^4 [-\tfrac{15}{8} + \tfrac{15}{4}s'^2 - \tfrac{105}{64}s'^4 + (-\tfrac{3}{4} + \tfrac{9}{4}s'^2 - \tfrac{27}{16}s'^4)\eta' \\
&\quad + (\tfrac{3}{8} - \tfrac{3}{8}s'^2 - \tfrac{15}{64}s'^4)\eta'^2 + (\tfrac{21}{16} - \tfrac{45}{32}s'^2)s'^2 e'^2 \cos 2g' \\
&\quad - (\tfrac{1}{2} - \tfrac{3}{2}s'^2 + \tfrac{9}{8}s'^4)\beta' e' \cos f' - (\tfrac{1}{4} - \tfrac{3}{4}s'^2 + \tfrac{9}{16}s'^4)\beta' e'^2 \cos 2f'].
\end{aligned}$$

The induction continues with evaluating the provisional element $\tilde{M}_{0,2} = \tilde{M}_{1,1} + (M_{0,1}; W_1)$. Because the term $M_{0,1}$ is already normalized, the Poisson bracket is more readily calculated in the Delaunay than in the Whittaker chart. Only the two partial derivatives

$$G' \frac{\partial M_{0,1}}{\partial L'} = -3 \frac{G'^2}{a'^2} \left(\frac{\alpha}{p'}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s'^2),$$

$$\frac{1}{G'} \frac{\partial W_1}{\partial l'} = \left(\frac{\alpha}{p'}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s'^2)\left(\frac{a'^2}{r'^2}\eta' - 1\right)$$

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are needed to find that

$$\tilde{M}_{0,2} = \tilde{M}_{1,1} - \left(G' \frac{\partial M_{0,1}}{\partial L'} \right) \left(\frac{1}{G'} \frac{\partial W_1}{\partial l'} \right)$$

$$= -3n'^2 a'^2 \left(\frac{\alpha}{p'} \right)^4 (\tfrac{1}{2} - \tfrac{3}{4} s'^2)^2 \eta'^2$$

$$+ \frac{G'^2}{r'^2} \left(\frac{\alpha}{p'} \right)^4 [-\tfrac{15}{8} + \tfrac{15}{4} s'^2 - \tfrac{105}{64} s'^4 + (\tfrac{3}{8} - \tfrac{3}{8} s'^2 - \tfrac{15}{64} s'^4) \eta'^2$$

$$+ (\tfrac{21}{16} - \tfrac{45}{32} s'^2) s'^2 e'^2 \cos 2g' - (\tfrac{1}{2} - \tfrac{3}{2} s'^2 + \tfrac{9}{8} s'^4) \beta' e' \cos f'$$

$$- (\tfrac{1}{4} - \tfrac{3}{4} s'^2 + \tfrac{9}{16} s'^4) \beta' e'^2 \cos 2f'].$$

The Lie equation at order 2

$$n' \frac{\partial W_2}{\partial l'} - n_e \frac{\partial W_2}{\partial h'} + M_{0,2} = \tilde{M}_{0,2}$$

is solved in the same way as that at order 1. The averaged Hamiltonian is

$$M_{0,2} = n' G' \left(\frac{\alpha}{p'} \right)^4 [-\tfrac{15}{8} + \tfrac{15}{4} s'^2 - \tfrac{105}{64} s'^4 + (-\tfrac{3}{4} + \tfrac{9}{4} s'^2 - \tfrac{27}{16} s'^4) \eta'$$

$$+ (\tfrac{3}{8} - \tfrac{3}{8} s'^2 - \tfrac{15}{64} s'^4) \eta'^2 + (\tfrac{21}{16} - \tfrac{45}{32} s'^2) s'^2 e'^2 \cos 2g'].$$

If s'^2 is replaced by $1 - c'^2$ in $M_{0,2}$, then one recovers term for term the second-order averaged Hamiltonian $F_2^* = -(-J_2)^2 M_{0,2}$ produced by Brouwer (see Brouwer and Clemence, 1961, p. 569). Straightforward quadratures will in turn yield the generator

$$W_2 = G' \left(\frac{\alpha}{p'} \right)^4 \{ \phi' [-\tfrac{15}{8} + \tfrac{15}{4} s'^2 - \tfrac{105}{64} s'^4$$

$$+ (\tfrac{3}{8} - \tfrac{3}{8} s'^2 - \tfrac{15}{64} s'^4) \eta'^2 + (\tfrac{21}{16} - \tfrac{45}{32} s'^2) s'^2 e'^2 \cos 2g']$$

$$- (\tfrac{1}{2} - \tfrac{3}{2} s'^2 + \tfrac{9}{8} s'^4) \beta' e' \sin f' - (\tfrac{1}{8} - \tfrac{3}{8} s'^2 + \tfrac{9}{32} s'^4) \beta' e'^2 \sin 2f' \}, \quad (53)$$

definitely shorter and simpler than the generator S_2 for the Poincaré transformation at order 2 obtained by Kozai (1962, Equation 3.2, p. 448). By applying the rule given by Henrard (1970, proposition 6), one may probably reconstruct the latter as the generator of the mapping which combines the elimination of the parallax and of the mean anomaly in a single operation.

With so simple a second-order generator as (53), it becomes possible to extend Brouwer's theory by computer to the third order – without omitting terms as was deliberately done by Kozai in an attempt to alleviate the massive complexity in the intermediate results.

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7. Natural Intermediaries

The elimination of the parallax is an instance of a general technique which, in retrospect, may be related to an approach taken by Sterne, Garfinkel and others. Consider the Hamiltonians $J \equiv J(p, P; \varepsilon)$ and

$$H \equiv H(q, Q; \varepsilon) = H_0(q, Q) + \varepsilon H_1(q, Q) + O(\varepsilon^2); \tag{54}$$

we say that J is a *natural intermediary* of H if (i) J is integrable in the map (p, P) about $\varepsilon = 0$; (ii) there exists a canonical transformation $\phi : (q, Q; \varepsilon) \to (p, P)$ such that

$$H(q[p, P; \varepsilon), Q(p, P, \varepsilon); \varepsilon] = J(p, P; \varepsilon) + O(\varepsilon^n) \quad (n \geqslant 2)$$

What the definition says in practical terms is that the dynamical system J may be solved in the phase space (p, P) and that its solution constitutes an approximation to order 1 at least of the solution for the dynamical system H in the map (p, P). Replacing the Hamiltonian H by its natural intermediary J amounts to pushing down the perturbation from the first order to a higher order. Whenever it is feasible, the downgrading is a very effective way of advancing the solution of the original problem. But the bait comes wrapped around a barb, not the least part of it consisting in a swapping of primary models.

Take for instance the dynamical system

$$M = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r} - \frac{1}{2}\frac{\mu\varepsilon}{r^2} + O(\varepsilon^2). \tag{55}$$

In the classical approach, one thinks of it as a Keplerian orbit continuously deformed and rotated by the perturbations in ε and ε^2. On the other hand, the torsion described in Section 3 sets the Keplerian Hamiltonian

$$J = \frac{1}{2}\left(R'^2 + \frac{\Theta'^2}{r'^2}\right) - \frac{\mu}{r'}$$

as a natural intermediary of (54). The model is now changed: the torsion gets the satellite to travel a rosette at the command of a phantom; the intermediary J puts the phantom on a Keplerian orbit; the perturbations bear on the phantom's motions. Questions about how the perturbations affect the perigee or the inclination of the satellite are no longer pertinent. The introduction of the pair (J, ϕ) as natural intermediary has replaced the old model by a new *gedanken-mechanism.*

In the main problem for the theory of artificial satellites, the elimination of the parallax may be used to define a chain of natural intermediaries. Their definitions follow easily from a new version of the reduced Hamiltonian (51) in which the quantities C and S are replaced by the more familiar elements e and g. After the substitutions (14) have been made, the Hamiltonian becomes

$$M = M_{0,0} + \frac{\Theta^2}{r^2} \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!}\left(\frac{\alpha}{p}\right)^{2n} \sum_{0 \leqslant j \leqslant n/2} e^{2j} \sum_{0 \leqslant k \leqslant j} M^*_{n,j,k} s^{2k} \cos 2kg, \tag{56}$$

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TABLE II
Inclination functions for the main problem in the theory of artificial satellites

$M^*_{1,0,0} =$	$\frac{1}{2}$	$-\frac{3}{4}s^2$			
$M^*_{2,0,0} =$	$-\frac{5}{4}$	$+\frac{21}{8}s^2$	$-\frac{21}{16}s^4$		
$M^*_{2,1,0} =$	$-\frac{3}{8}$	$+\frac{3}{8}s^2$	$+\frac{15}{64}s^4$		
$M^*_{2,1,1} =$	$\frac{21}{16}$	$-\frac{45}{32}s^2$			
$M^*_{3,0,0} =$	$\frac{39}{4}$	$-\frac{567}{16}s^2$	$+\frac{2961}{64}s^4$	$-\frac{315}{16}s^6$	
$M^*_{3,1,0} =$	$\frac{87}{16}$	$-\frac{837}{32}s^2$	$+\frac{6813}{128}s^4$	$-\frac{8145}{256}s^6$	
$M^*_{3,1,1} =$	$-\frac{9}{16}$	$-\frac{117}{32}s^2$	$+\frac{2565}{512}s^4$		
$M^*_{4,0,0} =$	$-\frac{501}{4}$	$+\frac{18\,909}{32}s^2$	$-\frac{131\,157}{128}s^4$	$+\frac{50\,049}{64}s^6$	$-\frac{13\,815}{64}s^8$
$M^*_{4,1,0} =$	$-\frac{3633}{32}$	$+\frac{11\,961}{32}s^2$	$+\frac{22\,509}{256}s^4$	$-\frac{123\,309}{128}s^6$	$+\frac{2\,596\,275}{4096}s^8$
$M^*_{4,1,1} =$	$\frac{40\,545}{64}$	$-\frac{300\,525}{128}s^2$	$+\frac{2\,956\,191}{1024}s^4$	$-\frac{2\,360\,115}{2048}s^6$	
$M^*_{4,2,0} =$	$-\frac{783}{128}$	$+\frac{13\,905}{256}s^2$	$-\frac{26\,541}{128}s^4$	$+\frac{277\,425}{1024}s^6$	$-\frac{1\,781\,595}{16384}s^8$
$M^*_{4,2,1} =$	$-\frac{567}{32}$	$+\frac{7533}{256}s^2$	$+\frac{50\,409}{2048}s^4$	$-\frac{136\,215}{4096}s^6$	
$M^*_{4,2,2} =$	$\frac{37\,611}{1024}$	$-\frac{10\,665}{128}s^2$	$+\frac{384\,345}{8192}s^4$		

the coefficients being the polynomials in s^2 listed in Table II. Incidentally, in (56), the pairs (e, g) and (s^2, g) both present the d'Alembert characteristic. This is the reason why the inclination polynomials $M^*_{n,j,k}$ are given in powers of s^2, as suggested by Aksnes, rather than in powers of c^2 as do Brouwer and Kozai, thereby reducing the total number of terms by about 10%.

The first two terms of (56), namely

$$R_0 = M_{0,0} + \varepsilon \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 M^*_{1,0,0}, \tag{57}$$

make a quasi-Keplerian approximation that is separable, hence (57) constitutes a natural intermediary. The canonical transformation that produces it is precisely the elimination of the parallax.

Responding to Garfinkel's suggestion of incorporating as much as possible of the perturbations in the intermediary, one could adjoin to (57) all terms of (56) beyond the first order which are independent of either the eccentricity or the argument of perigee. The resulting Hamiltonian

$$R_1 = M_{0,0} + \frac{\Theta^2}{r^2} \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!}\left(\frac{\alpha}{p}\right)^{2n} M^*_{n,0,0} \tag{58}$$

is also a natural intermediary. Indeed, it is separable since the torsion defined by

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the generator

$$T = r'R + v'N + \theta'\Theta\left[1 + \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!}\left(\frac{\alpha}{p}\right)^{2n} M^*_{n,0,0}\right]^{1/2}$$

reduces it to a Keplerian system. That the torsion defines the new variables by implicit equations may cause difficulties in an automated development for the main problem. An extension of Lagrange's inversion theorem (Deprit, 1979) allows, in principle, for recursive explicit expansions of the new variables and functions thereof. But the technique was not yet available at the time the author proposed an automated procedure to normalize R_1 in a Delaunay chart (Deprit, 1978).

A natural intermediary need not be separable; integrability is sufficient. Therefore, one may go another step toward meeting Garfinkel's goal, by adding to (58) all terms depending on the eccentricity e, leaving in the perturbation only those which depend explicitly on the argument of perigee. The angles θ and v being ignorable in the Hamiltonian

$$R_2 = M_{0,0} + \frac{\Theta^2}{r^2} \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!}\left(\frac{\alpha}{p}\right)^{2n} \sum_{0 \leqslant j \leqslant n/2} e^{2j} M^*_{n,j,0}, \tag{59}$$

the dynamical system (59) has only one degree of freedom and so is integrable. It makes a natural intermediary for the main problem in the theory of artificial satellites. How R_2 is normalized in a Delaunay chart will be explained in a forthcoming publication (Deprit, 1981).

8. Naturalization of Common Intermediaries

A natural intermediary is not an intermediary in the common sense of the word. We propose to elucidate the difference between the two concepts; then we show how the common intermediaries, in polar variables proposed for the main problem in the theory of artificial satellites, can be reset as natural intermediaries. This is a positive step, for natural intermediaries hasten the solution of the main problem more effectively than common intermediaries.

A dynamical system with Hamiltonian $J \equiv J(q, Q; \varepsilon)$ is called a (*common*) *intermediary* of (54) if

(i) J is of the form

$$J = H_0 + \varepsilon J_1(q, Q) + O(\varepsilon^2); \tag{60}$$

(ii) J is integrable;

(iii) The difference $H - J$ expressed in the angle-action variables (a, A) of J is conditionally periodic in the angles a.

The averaging algorithms of celestial mechanics are not applicable to (54) unless the principal term H_0 is an intermediary of H. The search for intermediaries of

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H other than H_0 stems from the expectation that, if J contains at least part of the first-order terms of H, then the solutions of J are 'better approximations' of the solutions of H than are the solutions given by the principal term H_0. This principle is not a theorem: what is meant by a 'better approximation' is not even formulated! Nevertheless, it is generally accepted as a physical mode of operation and, as such, is often a fruitful source of 'faster' numerical schemes and 'simpler' algebraic algorithms.

Garfinkel's definition of an intermediary is dominated by the requirement that the Hamilton–Jacobi equation for J be separable in the chart (q, Q) itself (Garfinkel and Aksnes, 1970). If I read Garfinkel correctly, the sole purpose for the restriction is to arrive immediately at a set of angle-action variables for J. Separation of the coordinates in the equation yields a complete solution $S(q, U, t;\varepsilon)$ by quadratures. The latter defines a time-dependent canonical transformation $\sigma:(q, Q, t) \to (u, U)$ by the implicit equations

$$Q = \frac{\partial S}{\partial q}, \qquad u = \frac{\partial S}{\partial U}.$$

The mapping σ drives the Hamiltonian J to equilibrium; therefore the momenta U are integrals of J and, because the transformation σ is canonical, they are in involution. According to Liouville's theorem, J is thus integrable. In the elementary context of a separable system, the problem of finding a set of action-angle variables for J may sometimes be solved in a simple way. One builds a time-dependent canonical transformation $\tau:(u, U, t) \to (a, A)$ whose remainder R depends on the momenta A only, and is such that the composition $\phi = \tau \circ \sigma:(q, Q) \to (a, A)$ is independent of the time (see e.g. Smart, 1953, pp. 167–169). Evidently $J[q(a, A), Q(a, A)] = R[u(a, A), U(a, A)] = R(A)$.

The restriction put by Garfinkel is not essential. A dynamical system with Hamiltonian $J(q, Q)$ is separable if there exists a canonical transformation $(q, Q) \to (p, P)$ such that the Hamilton–Jacobi equation of J is separable in the chart (p, P). In agreement with this definition, the natural intermediaries R_0 and R_1 are separable, although their Hamilton–Jacobi equations are not separable in the original Whittaker chart.

The condition that the residual perturbation $H - J$ be multiply-periodic in the angles a implies Garfinkel's requirement that the intermediary J 'cover all the first-order secular variations' of H. Indeed, apply Poincaré's method to the transformed $H(a, A;\varepsilon) = J(A;\varepsilon) + \varepsilon R(a, A) + O(\varepsilon^2)$ of H in the chart (a, A). At order 1, it will produce an infinitesimal contact transformation

$$a = a' + \varepsilon \frac{\partial W}{\partial A'}, \qquad A = A' - \varepsilon \frac{\partial W}{\partial a'}, \tag{61}$$

whose generator is determined by the partial differential formula $(J;W) + R' = R$. By assumption, R is periodic in the angles a'; hence one may select R' as the average

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of R over the angles a' while W, being determined by quadratures from $R - R'$, turns out to be a periodic function of the angles a'. To the first order, the Hamiltonian H becomes

$$H = J(A';\varepsilon) + \varepsilon R'(A') + O(\varepsilon^2) \tag{62}$$

in the chart (a', A'), which implies that the variables (a', A') make up a set of angles and actions for H. So Equations (61) of the infinitesimal contact transformation may be read as meaning that, save for the addition of first-order periodic corrections, the angle-action variables of J 'cover' the secular variations in the angle-action variables of H.

For Garfinkel, the search for 'effective' intermediaries in the main problem is confined to the original Whittaker chart. It rests on the proposition that quasi-Staeckel systems

$$J = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) + V_1(r, \Theta, N) + \frac{\Theta^2}{r^2} V_2(\theta, \Theta, N)$$

are integrable; partial clarification of this statement will be given elsewhere (Deprit and Richardson, 1981). The search consists in adjusting the components V_1 and V_2 to achieve specified effects at the first order and possibly beyond it in the course of a Poincaré–Lindstedt normalization of the main problem based on the intermediary. Depending on whether $V_2 = 0$ or $V_2 = \varepsilon\frac{3}{4}(\alpha/p)^2 \cos 2\theta$, the intermediary J is said to be *radial* or *zonal*.

For the main problem in the theory of artificial satellites, the first intermediary answering Garfinkel's requirements, namely

$$J_s = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r} + \varepsilon\frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2\left[(\tfrac{1}{2} - \tfrac{3}{4}s^2)\frac{p}{r} + \frac{3}{4}s^2\cos 2\theta\right], \tag{63}$$

was proposed by Sterne (1957); it was immediately followed by

$$J_g = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r} + \varepsilon\frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2\left[3(\tfrac{1}{2} - \tfrac{3}{4}s^2)\left(1 - \frac{r}{p}\eta\right) + \frac{3}{4}s^2\cos 2\theta\right] \tag{64}$$

due to Garfinkel (1958 and 1959, revised in 1964). Both are zonal intermediaries. Cid–Palacios (1969) came with the first radial intermediary

$$J_c = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r} + \varepsilon\frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2(\tfrac{1}{2} - \tfrac{3}{4}s^2)\frac{p}{r}. \tag{65}$$

Equation (62) brings to notice that the 'effectiveness' of an intermediary is enhanced if it is such that $R'(A') = 0$, i.e. if the residual perturbation $H(q, Q) - J(q, Q)$ averages to zero over the angle variables of the intermediary. As was discovered by Aksnes (1965 and 1966), the zonal intermediary

$$Z_0 = \frac{1}{2}\left(R^2 + \frac{\Theta^2}{r^2}\right) - \frac{\mu}{r} + \varepsilon\frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2[(\tfrac{1}{2} - \tfrac{3}{4}s^2) + \tfrac{3}{4}s^2\cos 2\theta] \tag{66}$$

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enjoys this property in relation to the main problem in the theory of artificial satellites.

The most effective intermediaries would be separable Hamiltonians of the form $J = H_0 + \varepsilon H_1 + \varepsilon^2 J_2(q, Q) + O(\varepsilon^3)$. If the sum $H_0 + \varepsilon H_1$ itself was separable, there would be no reason to look for an intermediary at the first order. But it is conceivable that, by adding terms of order 2 and higher to the non-separable sum $H_0 + \varepsilon H_1$, one could arrive at a separable Hamiltonian. A case in point is the *spheroidal* intermediary (Vinti, 1959; Kislik, 1961) for the main problem in the theory of artificial satellites. On the pattern of what was accomplished by the elimination of the parallax, we propose yet another way of reaching for the best intermediaries, which is to prepare a canonical transformation $\phi:(q, Q, \varepsilon) \to (p, P)$ casting the Hamiltonian H into a series

$$H^*(p, P, \varepsilon) = H_0^*(p, P) + \sum_{n \geqslant 1} \frac{\varepsilon^n}{n!} H_n^*(p, P)$$

with the aim of making the sum $J(p, P) = H_0^* + \varepsilon H_1^*$ separable in the chart (p, P). The pair (ϕ, J) will then make a natural intermediary. It must be stressed that, in this approach, it is the main problem (not a surrogate!) which becomes separable at order 1, not in the old chart but in the new chart (p, P).

The common intermediaries (63)–(66) can all be recovered as natural intermediaries for the main problem in the theory of artificial satellites. The statement is proved by producing the infinitesimal contact transformation leading from the main problem to the intermediary, and this is done by solving a Lie equation of the form

$$L_0(W) = M - J, \tag{67}$$

using the techniques outlined in Section 2.

For the intermediary of Cid–Palacios, the right-hand member of (67) is

$$M - J_c = \frac{3}{4}\frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 \frac{p}{r} s^2 \cos 2\theta,$$

hence the generator

$$W_c = \Theta\left(\frac{\alpha}{p}\right)^2 s^2 \left[\tfrac{3}{8}\sin 2\theta + \frac{Cp}{\Theta}(\tfrac{3}{8}\sin\theta + \tfrac{1}{8}\sin 3\theta) + \frac{Sp}{\Theta}(\tfrac{3}{8}\cos\theta - \tfrac{1}{8}\cos 3\theta)\right]$$

which Cid–Palacios obtained as the first-order term in his von Zeipel transformation.

In the case of Sterne's Hamiltonian,

$$M - J_s = \frac{3}{4}\frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 \left(\frac{p}{r} - 1\right) s^2 \cos 2\theta,$$

$$W_s = \Theta\left(\frac{\alpha}{p}\right)^2 s^2 \left[\frac{Cp}{\Theta}(\tfrac{3}{8}\sin\theta + \tfrac{1}{8}\sin 3\theta) + \frac{Sp}{\Theta}(\tfrac{3}{8}\cos\theta - \tfrac{1}{8}\cos 3\theta)\right],$$

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while, for Aksnes' zonal intermediary,

$$M - Z_0 = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2\left(\frac{p}{r} - 1\right)[(\tfrac{1}{2} - \tfrac{3}{4}s^2) + \tfrac{3}{4}s^2 \cos 2\theta],$$

$$W_a = \Theta\left(\frac{\alpha}{p}\right)^2\left\{\frac{Cp}{\Theta}[(\tfrac{1}{2} - \tfrac{3}{8}s^2)\sin\theta + \tfrac{1}{8}s^2 \sin 3\theta]\right.$$

$$\left. - \frac{Sp}{\Theta}[(\tfrac{1}{2} - \tfrac{9}{8}s^2)\cos\theta + \tfrac{1}{8}s^2 \cos 3\theta]\right\}.$$

For Garfinkel's intermediary taken as the extension

$$J_g = Z_0 + \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2(\tfrac{1}{2} - \tfrac{3}{4}s^2)\left(2 - 3\frac{r}{p}\eta\right)$$

of Aksnes' zonal Hamiltonian, the generator of the infinitesimal contact transformation which sets it as a natural intermediary of the main problem is of the form $W_g = W_a + W_g^\dagger$, the additional term being a solution of the Lie equation

$$L_0(W) = \left(\frac{\alpha}{p}\right)^2(\tfrac{1}{2} - \tfrac{3}{4}s^2)\left(3\frac{\mu}{r}\eta - 2\frac{\Theta^2}{r^2}\right).$$

But, in the Delaunay chart,

$$L_0(E) = \frac{\partial E}{\partial l}\frac{\partial M_0}{\partial L} = n\frac{a}{r},$$

hence the solution

$$W_g^\dagger = \Theta\left(\frac{\alpha}{p}\right)^2(\tfrac{1}{2} - \tfrac{3}{4}s^2)(3E - 2\theta).$$

There are many ways in which the infinitesimal contact transformation leading to a natural intermediary may be extended into a canonical transformation. Not to be overlooked is the straightforward continuation by the exponential mapping

$$q = p + \sum_{n \geqslant 1}\frac{\varepsilon^n}{n!}L_W^n(p) = \exp(\varepsilon L_W)(p),$$

$$Q = P + \sum_{n \geqslant 1}\frac{\varepsilon^n}{n!}L_W^n(P) = \exp(\varepsilon L_W)(P),$$

the differential operators being defined inductively by the sequence

$$L_W(F) = L_W^n[L_W^{n-1}(F)] \quad (n \geqslant 2)$$

starting with the derivations $L_W^0(F) = F$ and $L_W(F) = (F\,;W)$.

For comparison, we completed to order 4 the exponential continuation for the

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infinitesimal contact transformations which naturalize the intermediaries of Sterne and Aksnes. The continuations yield the series

$$M = J_s + \tfrac{1}{2}\varepsilon^2 \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^4 s^2 S_{2,0} + O(\varepsilon^3),$$

$$M = Z_0 + \tfrac{1}{2}\varepsilon^2 \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^4 Z_{2,0} + O(\varepsilon^3).$$

Only the parts of order 2 are presented here, those of orders 3 and 4 being too long for a conventional print. In the case of Sterne,

$$\begin{aligned}
S_{2,0} = {} & -\tfrac{33}{32}s^2 + (\tfrac{3}{4} - \tfrac{9}{8}s^2)\cos 2\theta + \tfrac{63}{32}s^2 \cos 4\theta \\
& + \frac{Cp}{\Theta}[\tfrac{9}{8}s^2 \cos\theta + (\tfrac{3}{2} - \tfrac{21}{16}s^2)\cos 3\theta + \tfrac{15}{16}s^2 \cos 5\theta] \\
& + \frac{Sp}{\Theta}[-(6 - \tfrac{63}{8}s^2)\sin\theta + (\tfrac{3}{2} - \tfrac{39}{16}s^2)\sin 3\theta + \tfrac{15}{16}s^2 \sin 5\theta] \\
& + \frac{C^2p^2}{\Theta^2}[(\tfrac{9}{16} - \tfrac{21}{64}s^2) + (\tfrac{9}{8} - \tfrac{141}{128}s^2)\cos 2\theta + (\tfrac{9}{16} - \tfrac{45}{64}s^2)\cos 4\theta \\
& - \tfrac{9}{128}s^2 \cos 6\theta] + \frac{CSp^2}{\Theta^2}[-(\tfrac{3}{2} - \tfrac{153}{64}s^2)\sin 2\theta + (\tfrac{9}{8} - \tfrac{21}{16}s^2)\sin 4\theta \\
& + \tfrac{9}{64}s^2 \sin 6\theta] + \frac{S^2p^2}{\Theta^2}[-(\tfrac{33}{16} - \tfrac{159}{64}s^2) + (\tfrac{21}{8} - \tfrac{387}{128}s^2)\cos 2\theta \\
& - (\tfrac{9}{16} - \tfrac{39}{64}s^2)\cos 4\theta + \tfrac{9}{128}s^2 \cos 6\theta],
\end{aligned}$$

and, for Aksnes' zonal intermediary,

$$\begin{aligned}
Z_{2,0} = {} & -\tfrac{5}{4} + \tfrac{15}{4}s^2 - \tfrac{123}{32}s^4 - (\tfrac{5}{4} - \tfrac{15}{8}s^2)s^2 \cos 2\theta + \tfrac{63}{32}s^4 \cos 4\theta \\
& + \frac{Cp}{\Theta}[-(2 - \tfrac{35}{4}s^2 + \tfrac{51}{8}s^4)\cos\theta - (\tfrac{15}{4} - \tfrac{87}{16}s^2)s^2 \cos 3\theta \\
& + \tfrac{15}{16}s^4 \cos 5\theta] + \frac{Sp}{\Theta}[-(2 + \tfrac{11}{4}s^2 - \tfrac{51}{8}s^4)\sin\theta - (\tfrac{15}{4} - \tfrac{69}{16}s^2)s^2 \sin 3\theta \\
& + \tfrac{15}{16}s^4 \sin 5\theta] + \frac{C^2p^2}{\Theta^2}[-\tfrac{3}{8} + \tfrac{27}{16}s^2 - \tfrac{75}{64}s^4 - (\tfrac{3}{8} - \tfrac{3}{4}s^2 + \tfrac{9}{128}s^4)\cos 2\theta \\
& - (\tfrac{15}{16} - \tfrac{75}{64}s^2)s^2 \cos 4\theta + \tfrac{9}{128}s^4 \cos 6\theta] + \frac{CSp^2}{\Theta^2}[-(\tfrac{3}{4} - \tfrac{3}{4}s^2 \\
& - \tfrac{45}{64}s^4)\sin 2\theta - (\tfrac{15}{8} - \tfrac{39}{16}s^2)s^2 \sin 4\theta + \tfrac{9}{64}s^4 \sin 6\theta] \\
& + \frac{S^2p^2}{\Theta^2}[-\tfrac{3}{8} - \tfrac{15}{16}s^2 + \tfrac{105}{64}s^4 + (\tfrac{3}{8} - \tfrac{39}{128}s^4)\cos 2\theta \\
& + (\tfrac{15}{16} - \tfrac{81}{64}s^2)s^2 \cos 4\theta - \tfrac{9}{128}s^4 \cos 6\theta].
\end{aligned}$$

Symmetric though they are, these expressions are not simple enough. Undeniably

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they squander most of the advantages gained from having assimilated the whole first-order perturbation into the intermediary by naturalization. From a practical standpoint, the bulkiness of the result implies that the exponential continuation of the naturalizing infinitesimal contact transformation is too loose, that it burdens the main problem with too many periodic fluctuations around the intermediary's model. The continuation should remove these non-essential terms as it proceeds from order to order.

We proposed that, past order 1, the continuation be implemented as an elimination of the parallax, in that it converts parallactic factors α/r into ratios α/p and removes the terms explicitly periodic in θ. Acting on that suggestion, Dr Alfriend at the Naval Research Laboratory conducted the naturalization for the zonal intermediary (66) to order 3. The result is a series

$$M = Z_0 + \frac{\Theta^2}{r^2} \sum_{n \geqslant 2} \frac{\varepsilon^n}{n!} \left(\frac{\alpha}{p}\right)^{2n} \left[\sum_{0 \leqslant j \leqslant n/2} \sum_{0 \leqslant k \leqslant j} \left(\frac{Sp}{\Theta}\right)^{2k} \left(\frac{Cp}{\Theta}\right)^{2(j-k)} Z_{n,j,k} \right]$$

with the coefficients

$$Z_{2,0,0} = -\tfrac{5}{4} + \tfrac{15}{4} s^2 - \tfrac{123}{32} s^4,$$

$$Z_{2,1,0} = -\tfrac{3}{8} + \tfrac{27}{16} s^2 - \tfrac{75}{64} s^4 (= M_{2,1,0}),$$

$$Z_{2,0,1} = -\tfrac{3}{8} - \tfrac{15}{16} s^2 + \tfrac{105}{64} s^4 (= M_{2,0,1}),$$

$$Z_{3,0,0} = \tfrac{39}{4} - 27 s^2 + \tfrac{117}{32} s^4 + \tfrac{585}{32} s^6,$$

$$Z_{3,1,0} = \tfrac{87}{16} - \tfrac{1737}{64} s^2 + \tfrac{1647}{32} s^4 - \tfrac{225}{8} s^6,$$

$$Z_{3,0,1} = \tfrac{87}{16} - \tfrac{1611}{64} s^2 + \tfrac{927}{16} s^4 - \tfrac{2475}{64} s^6$$

comparable in size and form to the corresponding polynomials mentioned in Table I.

Here again Garfinkel's proposal to make the intermediary as inclusive as possible invites us to adjoin selected contributions of order 2 and 3 to the basic zonal intermediary Z_0. To this end, Alfriend's results are expressed in terms of e and g rather than C and S so that the main problem becomes the series

$$M = Z_0 + \frac{\Theta^2}{r^2} \sum_{n \geqslant 2} \frac{\varepsilon^n}{n!} \left(\frac{\alpha}{p}\right)^{2n} \sum_{0 \leqslant j \leqslant n/2} e^{2j} \sum_{0 \leqslant k \leqslant j} Z^*_{n,j,k} s^{2k} \cos 2kg$$

with the inclination polynomials

$$Z^*_{2,0,0} = -\tfrac{5}{4} + \tfrac{15}{4} s^2 - \tfrac{123}{32} s^4 (= Z_{2,0,0}),$$

$$Z^*_{2,1,0} = -\tfrac{3}{8} + \tfrac{3}{8} s^2 + \tfrac{15}{64} s^4 (= M^*_{2,1,0}),$$

$$Z^*_{2,1,1} = \tfrac{21}{16} - \tfrac{45}{32} s^2 (= M^*_{2,1,1}),$$

$$Z^*_{3,0,0} = \tfrac{39}{4} - 27 s^2 + \tfrac{117}{32} s^4 + \tfrac{585}{32} s^6 (= Z_{3,0,0}),$$

$$Z^*_{3,1,0} = \tfrac{87}{16} - \tfrac{837}{32} s^2 + \tfrac{2001}{64} s^4 - \tfrac{4275}{128} s^6,$$

$$Z^*_{3,1,1} = -\tfrac{63}{64} - \tfrac{207}{64} s^2 - \tfrac{675}{128} s^4.$$

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The natural intermediary

$$Z_1 = Z_0 + \frac{\Theta^2}{r^2} \sum_{n \geqslant 2} \frac{\varepsilon^n}{n!} \left(\frac{\alpha}{p}\right)^{2n} Z^*_{n,0,0} \tag{68}$$

is separable, whereas the natural intermediary

$$Z_2 = Z_1 + \frac{\Theta^2}{r^2} \sum_{n \geqslant 2} \frac{\varepsilon^n}{n!} \left(\frac{\alpha}{p}\right)^{2n} \sum_{1 \leqslant j \leqslant n/2} e^{2j} Z^*_{n,j,0} \tag{69}$$

TABLE III

Classification of the natural intermediaries in polar variables for the main problem

$$M_0 = \tfrac{1}{2}\left(R^2 + \frac{\theta^2}{r^2}\right) - \frac{\mu}{r}$$

$$\longrightarrow R_0 = M_0 + \varepsilon \frac{\theta^2}{r^2}\left(\frac{\alpha}{r}\right)^2 \left(\tfrac{1}{2} - \tfrac{3}{4}s^2\right)$$

$$\longrightarrow R_1 \longrightarrow R_2$$

$$\longrightarrow J_c = R_0 + \varepsilon \frac{\theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 \left(\tfrac{1}{2} - \tfrac{3}{4}s^2\right)\left(\frac{p}{r} - 1\right)$$

$$\longrightarrow Z_0 = R_0 + \tfrac{3}{4}\varepsilon \frac{\theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 s^2 \cos 2\theta$$

$$\longrightarrow Z_1 \longrightarrow Z_2$$

$$\longrightarrow J_s = Z_0 + \varepsilon \frac{\theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 \left(\tfrac{1}{2} - \tfrac{3}{4}s^2\right)\left(\frac{p}{r} - 1\right)$$

$$\longrightarrow J_g = Z_0 + \varepsilon \frac{\theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 \left(\tfrac{1}{2} - \tfrac{3}{4}s^2\right)\left(2 - 3\frac{r}{p}\eta\right)$$

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is integrable but not separable. Both may be more effective than Z_0. There are indications that this is the case. The first transformation by which Aksnes (1969, 1970) processes the residual perturbations $M - Z_0$, although it is announced as an averaging over the mean anomaly, is precisely the elimination of the parallax extending Z_0 into Z_2.

Now that all known intermediaries in polar variables have been shown to share the capital property of being natural intermediaries to the main problem in the theory of artificial satellites, is there a reason to favour one rather than the others? By the 'criteria of merit' proposed by Garfinkel and Aksnes (1970), the radial intermediaries rate higher than the zonal ones since the separation of variables in their Hamilton–Jacobi equation does not involve elliptic integral for the argument of latitude. By the same criteria, among the radial intermediaries, the Hamiltonian R_0 is especially commendable for '1. its accuracy, 2. its simplicity, and 3. the ease of calculating higher order perturbations'.

In Table III, the natural intermediaries in Whittaker variables for the main problem have been ordered in a genealogical tree by increasing complexity of their components V_1 and V_2. By duality, the tree constitutes an ordering by increasing simplicity in the generators and the equations of their naturalizing canonical transformations. Which intermediary one picks on the tree is a matter of personal taste, commanded to a large extent by the software context. After all, no one said it with more glee than Prolix (Goscinny and Uderzo, 1972): *Simplicity is difficult to program, but recognizable when it occurs.*

9. The Radial Intermediary

The parallax having been eliminated, one may pursue the solution of the main problem by adopting the Keplerian principal part M_0 as the intermediary, thus viewing the orbit as a kind of envelope of a family of ellipses in continuing deformation. As was indicated in Section 6, an averaging over the mean anomaly will then reproduce the Hamiltonian in the perigee elements (g, G) obtained by Brouwer and Kozai. In a subsequent averaging over the argument of perigee, the Poincaré-Lindstedt method breaks down formally at the singular inclinations $I_c \bmod \pi = \tan^{-1} 2$; away from these singularities, it produces first-order long period variations in g.

In sharp contrast with Brouwer's reference ellipse stands the model based on the radial intermediary R_0 of Equation (57).

As a dynamical system, R_0 is integrable: the coordinates θ and v are ignorable, hence the momenta Θ and N are integrals. The intermediary may be reduced to a dynamical system with only one degree of freedom in (r, R); integration of the differential equations

$$\dot{r} = \frac{\partial R_0}{\partial R} = R, \tag{70$_1$}$$

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$$\dot{R} = -\frac{\partial R_0}{\partial r} = \frac{\Theta^2}{r^3}\left[1 + \varepsilon\left(\frac{\alpha}{p}\right)^2 (1 - \tfrac{3}{2}s^2)\right] - \frac{\mu}{r^2} \tag{70$_2$}$$

is to be followed by the quadratures

$$\dot{v} = \tfrac{3}{2}\varepsilon\frac{\Theta}{r^2}\left(\frac{\alpha}{p}\right)^2 c, \tag{70$_3$}$$

$$\dot{\theta} = \frac{\Theta}{r^2}\left[1 - \varepsilon\left(\frac{\alpha}{p}\right)^2 (\tfrac{5}{2} - 3s^2)\right]. \tag{70$_4$}$$

The differential system (70$_1$)–(70$_2$) may be solved in the conventional way. Setting

$$\Delta \equiv \Delta(\Theta, N, \varepsilon) = 1 + \varepsilon\left(\frac{\alpha}{p}\right)^2 (1 - \tfrac{3}{2}s^2),$$

$$\delta \equiv \delta(\Theta, N, \varepsilon) = \Delta^{-1/2}\left[1 + \tfrac{1}{2}\varepsilon\left(\frac{\alpha}{p}\right)^2 (1 - 6c^2)\right],$$

$$u = \Delta\frac{p}{r}, \qquad \psi = \delta^{-1}\theta,$$

one finds that the differential equation of the orbits is $\mathrm{d}^2u/\mathrm{d}\psi^2 + u = 1$. Its general solution $u = 1 + e\cos(\psi - g)$ introduces two arbitrary constants e and g in addition to the integrals Θ and N. Two more constants will appear in the quadratures (70$_3$) and (70$_4$).

On the basis of these results one might derive a set of canonical constants. A direct approach, however, is simpler: it consists in applying the torsion $(\theta, v, \Theta, N) \to (\psi, \phi, \Psi, \Phi)$ generated by the function

$$T \equiv T(\psi, \phi, \Theta, N) = \psi\Theta\Delta^{1/2} + \phi N \tag{71}$$

to convert R_0 into the Keplerian system

$$R_0 = \frac{1}{2}\left(R^2 + \frac{\Psi^2}{r^2}\right) - \frac{\mu}{r}. \tag{72}$$

The transformation is given by the implicit equations

$$\theta = \delta\Delta^{1/2}\psi, \qquad v = \phi + \tfrac{3}{2}\varepsilon\left(\frac{\alpha}{p}\right)^2 c\Delta^{-1/2}\psi, \tag{73}$$

$$\Psi = \Delta^{1/2}\Theta, \qquad \Phi = N.$$

It is now obvious that the orbits of R_0 are rosettes developed from Keplerian orbits by the geometric construction Newton has given in *Propositio XLIV*. To the elementary law given in (35) corresponds here the pair of equalities $\delta\psi/\theta = \Psi/\Theta = \Delta^{1/2}$. The variables (ϕ, ψ, Ψ, Φ) refer to the phantom satellite: the latter travels a conic fixed in dimensions and position and does it in accordance with Kepler's laws.

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At any given moment, the satellite and its phantom are at the same distance from the centre of the Earth and they have the same radial velocity. Intuitively speaking, the torsion has adjusted the parameters of the phantom's Keplerian motion to make superfluous the addition of short-period variations to the radial variables (r, R) of the satellite; for instance, the mean motion of the phantom is determined not by the principal part M_0 but by the *total* energy R_0. The phantom's perigee is fixed; the satellite's perigees form a polygonal contour on a fixed circle in the orbital plane. The torsion in θ translates in Brouwer's model into a uniform rotation of the perigee; the mean motion of perigee for Brouwer's orbit equals the rate of increase in the average of θ over the anomalistic period of the phantom.

So far, in contrast to the approximate solution restricted to bounded orbits as it was built by Brouwer, we have produced an *exact* solution for the radial intermediary valid everywhere in the phase space except on the submanifold $\psi = 0$. The generality may be an advantage in applications where the second-order perturbations are not significant and the time span is limited to a few revolutions. But if the radial intermediary is to provide reference orbits in a calculation of perturbations beyond the first order, it is desirable that the equations of the torsion be made explicit. The problem consists in solving the implicit equation $\Psi = \Theta\Delta^{1/2}$ and in substituting the solution $\Theta \equiv \Theta(\Psi, \Phi)$ in the equations for the angles θ and v. But we suggest that the torsion be reconstructed as a Lie transformation developed in the powers of ε. An exposition of the transformation in full falls outside the scope of this article; it is sufficient here to examine only the infinitesimal contact transformation leading from R_0 to the Keplerian system (73). Its generator is a solution of the Lie equation

$$L_0(W) = \varepsilon \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s^2);$$

hence, up to the addition of an arbitrary function in Θ and N,

$$W = \varepsilon\Theta\left(\frac{\alpha}{p}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}s^2)\theta \tag{74}$$

and the explicit equations of the torsion are

$$\theta = \psi\left[1 + \tfrac{3}{4}\varepsilon\left(\frac{\alpha}{q}\right)^2 (1 - 5\gamma^2)\right] + O(\varepsilon^2),$$

$$v = \phi + \tfrac{3}{2}\varepsilon\left(\frac{\alpha}{q}\right)^2 \gamma\psi + O(\varepsilon^2),$$

$$\Theta = \Psi\left[1 + \varepsilon\left(\frac{\alpha}{q}\right)^2 (\tfrac{1}{2} - \tfrac{3}{4}\sigma^2)\right] + O(\varepsilon^2),$$

$$N = \Phi,$$

the auxiliary functions q, γ and σ being given by the relations $\Psi^2 = \mu q$, $\gamma = \Phi/\Psi$ and

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$\sigma^2 = 1 - \gamma^2$. As one would expect, the values I_c for the inclination take on a special significance. At these inclinations, $\theta = \psi$ and the torsion in the orbital plan becomes the identity mapping. More precisely, the satellite's orbits are conics having a focus at the centre of the Earth, although the satellite's motion on the conical path does not obey Kepler's second law of areas.

Is it possible to build a solution to the main problem in the theory of artificial satellites on the model emanating from the radial intermediary? The solution would offer an alternative to the solutions built on the Keplerian model, not necessarily in the sense of providing a practical way of evaluating the second-order perturbations, but it would assist in separating the features of Brouwer's solution which belong intrinsically to the phase space of the main problem from those which are mere artifacts of the analytical procedure.

Conceivably the second-order perturbations have a two-fold effect. On the one hand they bring corrections on the rate of the torsion; on the other, they distort the Keplerian motion of the phantom's regulating mechanism. That much we know for certain: the perturbations cannot be targeted at the torsion exclusively. Indeed, let us attempt to build a Lie transformation with the aim of converting the Hamiltonian

$$M = M_0 + \varepsilon M_{1,0} + \tfrac{1}{2}\varepsilon^2 M_{2,0},$$

$$M_{1,0} = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^2 M_{1,0,0},$$

$$M_{2,0} = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^4\left(M_{2,0,0} + M_{2,1,0}\frac{C^2p^2}{\Theta^2} + M_{2,0,1}\frac{S^2p^2}{\Theta^2}\right)$$

into a Keplerian system. We have already obtained in (74) the first-order generator. Then, at the second order, we calculate that

$$\tilde{M}_{1,1} = \tilde{M}_{0,2} = \frac{\Theta^2}{r^2}\left(\frac{\alpha}{p}\right)^4\left(M^{\dagger}_{2,0,0} + M_{2,1,0}\frac{C^2p^2}{\Theta^2} + M_{2,0,1}\frac{S^2p^2}{\Theta^2}\right)$$

with the inclination polynomial

$$M^{\dagger}_{2,0,0} = -\tfrac{3}{4}s^2(1 - \tfrac{5}{4}s^2) = \tfrac{3}{16}s^2(1 - 5c^2).$$

According to the proposition proved in Section 1, the Lie equation $L_0(W_2) = \tilde{M}_{0,2}$ admits the solution

$$W_2 \equiv W_2(p, C, S, \theta) = \Theta\theta\left(\frac{\alpha}{p}\right)^4\left(M^{\dagger}_{2,0,0} + M_{2,1,0}\frac{C^2p^2}{\Theta^2} + M_{2,0,1}\frac{S^2p^2}{\Theta^2}\right).$$

The resulting equations for the radial elements are then of the form

$$r = r' + \varepsilon^2\frac{\partial W_2}{\partial R'} = r'[1 + \varepsilon^2\theta' k(e', s'^2, \theta', g')],$$

$$R = R' - \varepsilon^2\frac{\partial W_2}{\partial r'} = R'[1 + \varepsilon^2\theta' K(e', s'^2, \theta', g')],$$

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where k and K are multiply-periodic functions in θ' and g'. The presence of θ' as a factor makes this type of function worthless to an analysis of how the perturbations recondition the phase space of the radial intermediary. Nevertheless, the Lie transformation just built may have some practical value in contexts where one is interested in a solution accurate to the second order in ε^2 on a time span of a few anomalistic periods.

10. Conclusion

A case has been made for canonical transformations aimed not at generating integrals of a dynamical system but only at removing non-essential intricacies. Applied to the main problem in satellite theory, the elimination of the parallax has put the system in a state where the developments have become elementary at the second order and manageable without the help of a computer at the third order. Conceptually, it has offered a new perspective along which one can regroup various theories based on the notion of an intermediary and increase their effectiveness. Finally, the elimination of the parallax itself has produced a new class of intermediaries. It is not unreasonable to expect that the radial intermediary will contribute one day to our understanding of the phase flow in the main problem for the theory of artificial satellites.

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